Food As MedicinE for Diabetes Study Protocol

(**Brief Title**: FAME-D)

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STATEMENT OF COMPLIANCE

(1) [The trial will be carried out in accordance with International Council on Harmonisation Good Clinical Practice (ICH GCP) and the following:

- United States (US) Code of Federal Regulations (CFR) applicable to clinical studies (45 CFR Part 46, 21 CFR Part 50, 21 CFR Part 56, 21 CFR Part 312, and/or 21 CFR Part 812).

National Institutes of Health (NIH)-funded investigators and clinical trial site staff who are responsible for the conduct, management, or oversight of NIH-funded clinical trials have completed Human Subjects Protection and ICH GCP Training.

The protocol, informed consent form(s), recruitment materials, and all participant materials will be submitted to the IRB for review and approval. Approval of both the protocol and the consent form(s) must be obtained before any participant is consented. Any amendment to the protocol will require review and approval by the IRB before the changes are implemented to the study. All changes to the consent form(s) will be IRB approved; a determination will be made regarding whether a new consent needs to be obtained from participants who provided consent, using a previously approved consent form.

INVESTIGATOR'S SIGNATURE

The signature below constitutes the approval of this protocol and provides the necessary assurances that this study will be conducted according to all stipulations of the protocol, including all statements regarding confidentiality, and according to local legal and regulatory requirements and applicable US federal regulations and ICH guidelines, as described in the *Statement of Compliance* above.

Principal Investigator or Clinical Site Investigator:

Signed: _____ Date: _____

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1 PROTOCOL SUMMARY

1.1 SYNOPSIS

Trial Title	Food As MedicinE for Diabetes (FAME-D)
Grant Title:	Food is Medicine: A Randomized Clinical Trial of Medically Tailored Meals For Individuals with Type 2 Diabetes Mellitus and Food Insecurity
Grant Number:	R01DK125831

Study Description:

Food insecurity, "lack of access to enough food for an active, healthy life", affects over 20% of the 30 million Americans with diabetes. Food insecurity is more common in racial/ethnic minorities and those with lower socioeconomic status. Moreover, food insecurity is associated with worse diabetes control and increased complications, even after adjusting for other risk factors. Food insecurity is a major contributor to disparities in diabetes outcomes.

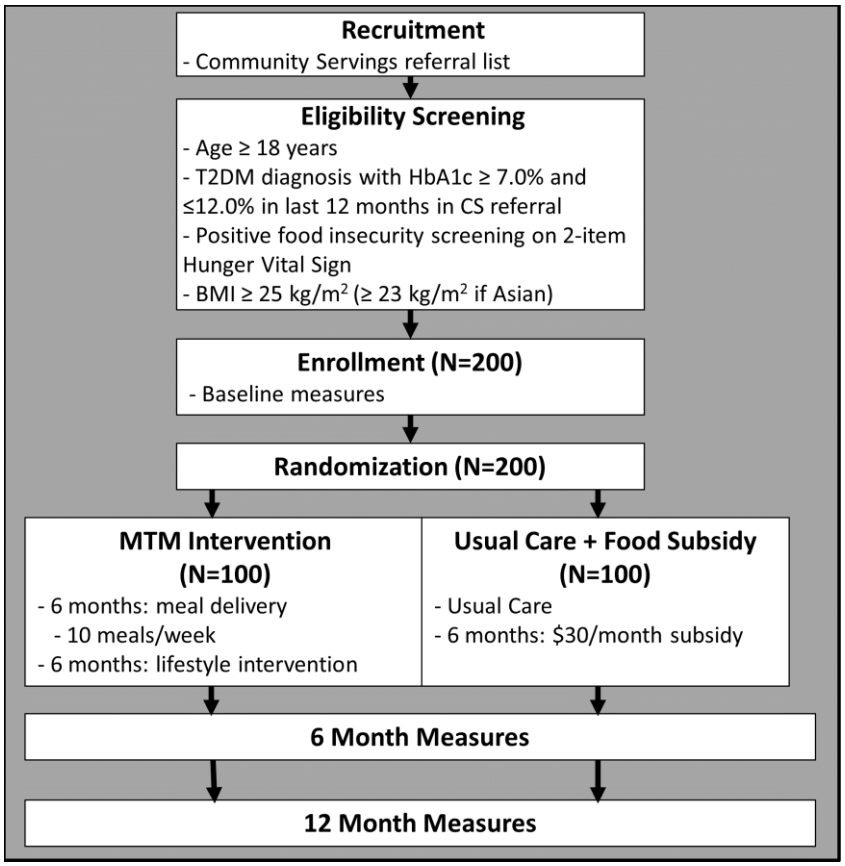
Medically tailored meal (MTM) delivery programs are a promising intervention for individuals with diabetes and food insecurity. MTM programs deliver fully prepared food, tailored by a registered dietitian nutritionist to the specific medical needs of the individual, and provide education to help optimize disease self-management. Food insecurity is typically addressed with food subsidies—offering additional financial resources that can be spent on food. While both food subsidies and MTM can increase healthy food access, MTM can help overcome other barriers to diabetes management including lack of time, ability, knowledge, and skills needed to prepare appropriate meals. The use of these programs is often called 'food as medicine', as the purpose is to provide exactly the foods needed to help prevent diabetes complications. MTM are receiving increasing public attention, but there are, as of yet, no full-scale trials to evaluate its effects on diabetes outcomes when compared with other food insecurity interventions.

Our research team has developed a medically tailored meal intervention that combines provision of healthy food, tailored to the specific nutrition needs of the individual, with an evidence-based lifestyle intervention that uses the period of meal delivery as springboard to improve diabetes self-management, both while receiving meals and after meal delivery is completed.

This study is an explanatory-focused randomized trial to assess a community-based medically tailored meals intervention (n=200). It will be conducted among diverse participants referred for medically tailored meals. Adults with type 2 diabetes, Hemoglobin A1c between 7.0% and 12.0%, and BMI ≥ 25 kg/m^2 (≥ 23 kg/m^2 for those with Asian ancestry) will be enrolled and randomly assigned to intervention or usual care + food subsidy. The intervention group will receive meal delivery and its attendant lifestyle intervention for 6 months, while the usual care + food subsidy group will receive a $40/month food subsidy, along with usual diabetes care.

Objectives*:	Aim 1 (QUAN): To determine the efficacy of a medically tailored meal intervention program for individuals with type 2 diabetes in a 6-month randomized clinical trial (n=200) Hypothesis 1a. Primary outcome. Hemoglobin A1c: We hypothesize that the MTM group will have 0.5% lower hemoglobin A1c at the end of treatment (6 months, primary analysis) and 6 months after treatment ends (12 months, secondary analysis), compared with the 'usual care + food subsidy' group. Hypothesis 1b. Patient-reported outcomes: We hypothesize that the MTM group will have less food insecurity, hypoglycemia, and diabetes distress, and greater diabetes self-efficacy, diet self-efficacy, and health-related quality of life, at the end of treatment (6 months) and 6 months after treatment ends (12 months), compared with the 'usual care + food subsidy' group. Aim 2 (qual and quan): To assess why, how, and under what circumstances MTM participants A) achieve improvements in T2DM outcomes and B) maintain these improvements after the intervention. Qualitative interviews with participants (N=100) at 6 and 12 months will examine cognitive ('why?'), procedural ('how?'), and contextual ('under what circumstances?') dimensions of their response, if any, to MTM. These will help identify potential mechanisms and moderators of treatment effects, such as race/ethnicity, and inform future modifications. Quantitative analyses will evaluate the 'how' by examining the mediating effect of changes in diet/weight, compensatory strategies, and psychological factors.
Endpoints*:	As outlined above under aims.
Phase* or Stage:	Phase II
Target Population	Individuals with type 2 diabetes, suboptimal glycemic control, and food insecurity.
Study Cohort	N=200 participants will be enrolled from among individuals referred to the Community Servings (CS) organization in Boston, MA.
Sites Enrolling Participants	CS is a Massachusetts-based 501(c)3 charitable not-for-profit organization that provides medically-tailored meals.
Treatment Regimens	Intervention: Delivery of medically tailored meals (10/week) along with monthly telephone-based lifestyle counseling sessions. Control: Provision of a $40/month food subsidy along with usual care. Each of the two regimens will be administered for 6 months.
Study Duration*:	42 months
Participant Duration:	12 months

1.2 SCHEMA


Recruitment
- Community Servings referral list
Eligibility Screening
- Age ≥ 18 years
- T2DM diagnosis with HbA1c ≥ 7.0% and ≤12.0% in last 12 months in CS referral
- Positive food insecurity screening on 2-item Hunger Vital Sign
- BMI ≥ 25 kg/m² (≥ 23 kg/m² if Asian)
Enrollment (N=200)
- Baseline measures
Randomization (N=200)
MTM Intervention (N=100)
- 6 months: meal delivery
- 10 meals/week
- 6 months: lifestyle intervention
Usual Care + Food Subsidy (N=100)
- Usual Care
- 6 months: $30/month subsidy
6 Month Measures
12 Month Measures

1.3 SCHEDULE OF ACTIVITIES

The schedule of study activities is described below.

Recruitment and Enrollment will occur during two phone calls.

- Eligibility phone call. Patients who are referred to Community Servings will be called to assess interest and eligibility. Anticipated call duration is 10-15 minutes. If interested and eligible, the potential participant will be sent the study consent form and asked to review before the consent phone call.
- Consent phone call. This call is planned for 1-2 weeks after the eligibility call. The consent process is expected to last 20 to 30 minutes. Because this is a low-risk study, verbal consent, over the telephone, will be used. Study staff will review the study in detail going over each key component of the consent form. If the potential participant agrees to participate, he/she will be provide verbal consent. After the consent phone call, those who agree to participate will continue on the call to complete baseline questionnaires over the telephone with a study team member. The study team member will collect these responses on-line (using REDCap). This part of the call is expected to last 45-60 minutes. After the completion of baseline questionnaires, the study team member will schedule a home-visiting nurse visit to collect data on participant height, weight, blood pressure, and perform a blood draw for a hemoglobin A1c (HbA1c). This nurse visit is expected to last 15-30 minutes, and can be completed the same day as the consent phone call, or later to accommodate the participant's schedule. The goal is for the visit to occur within 1 week.
- After the completion of the baseline questionnaires and nurse visit, the participant will be officially enrolled. Consenting individuals who are unwilling or unable to complete the baseline questionnaires and the nurse visit will not be enrolled. Each enrolled participant will be assigned by randomization to either the medically-tailored meal (MTM) intervention or to the food subsidy regimen, used as a comparison group.
- The MTM regimen will involve the following activities:
    - Set-up call to do medical tailoring of meals, make delivery arrangements, schedule lifestyle counseling sessions
    - Receive delivery of medically-tailored meals provided as 10 meals/week for 6 months
    - 6 (1 per month) telephone-based 1-on-1 lifestyle counseling sessions, with sessions lasting approximately 45 minutes
    - 6 (1 per month) check-in telephone calls, occurring 1-2 weeks after the lifestyle counseling session, to assess progress, follow-up on goals, and troubleshoot issues that arise
- The food subsidy regimen will involve the following activities:
    - Set-up call to go over logistical details, how to use the food subsidy, and go-over a provided document emphasizing purchase of healthy foods, using WIC-eligible foods as a model of fruits and vegetables with no added sugar, salt, or fat. Also go over provided healthy recipes.
    - Receive, by mail 6 times (1 per month), a $40 food subsidy to be used at local grocery stores, restricted from purchase of alcohol or tobacco

- 6 (1 per month) telephone-based 1-on-1 check-in sessions, lasting approximately 45 minutes, troubleshooting issues with using the subsidy and assessing foods purchased.

- Data collection activities applicable to both treatment regimens are as follows:
  - At approximately 6 months after the start of the intervention complete telephone-based questionnaires (similar to baseline visit), followed by home nurse visit for collection of height, weight, blood pressure, and blood draw for HbA1c.
  - At approximately 12 months after start of the intervention, complete telephone-based questionnaires (similar to baseline visit), followed by home nurse visit for collection of height, weight, blood pressure, and blood draw for HbA1c.
  - If selected, complete, after 6 month assessments (allowable window 0 to 3 months after 6 month assessment), a telephone based, 1-on-1 semi-structured interview lasting approximately 30 minutes.
  - If selected, complete, after 12 month assessments (allowable window 0 to 3 months after 12 month assessment), a telephone based, 1-on-1 semi-structured interview lasting approximately 30 minutes.

2 INTRODUCTION

2.1 STUDY RATIONALE

The purpose of the study is to evaluate the efficacy of a medically tailored meal intervention designed to improve diabetes outcomes such as Hemoglobin A1c, and patient-reported outcomes such as diabetes distress, quality of life, and hypoglycemia.

2.2 BACKGROUND

Defining the Problem

Food Insecurity is common in individuals with type 2 diabetes mellitus. Food insecurity is defined as "lack of access to enough food for an active, healthy life."[1] In 2017, 11.1% of American households, over 37 million individuals, experienced food insecurity.[2] Food insecurity is twice as common in individuals with type 2 diabetes mellitus (T2DM) than the general US population,[3–6] occurring in 5.4 million of the approximately 30.3 million Americans with diabetes.[7,8] Food insecurity is 2 to 2.5 times more common in racial/ethnic minorities (particularly non-Hispanic black and Hispanic individuals) and those with lower socioeconomic status.[7,9]

Food insecurity is strongly associated with both worse glycemic control and other adverse T2DM outcomes. Food insecurity is associated with greater odds (OR 1.5, p = 0.03) of hemoglobin A1c (HbA1c) > 9.0%[3] and higher mean HbA1c (mean difference 0.6%, p<0.0001) in cross-sectional and longitudinal studies from our group.[10,11] These studies adjusted for potential confounders such as income, education, race/ethnicity, age, gender, and other health-related social needs (e.g., housing instability, cost-related

medication underuse).[10,11] Others report similar findings, including in path analyses that estimate a direct effect from food insecurity to HbA1c.[12–19] Food insecurity is strongly associated with T2DM complications, including chronic kidney disease, diabetic retinopathy, coronary heart disease and congestive heart failure, peripheral arterial disease, and stroke.[4,20–23] Food insecurity is associated with lower diabetes self-efficacy, more depressive symptoms and diabetes distress, and higher all-cause mortality.[24–28]

Food insecurity undermines standard approaches to reducing T2DM complications. Standard T2DM management includes dietary modification and use of glucose-lowering medications. Food insecurity makes dietary modification difficult as it incentivizes the consumption of (often heavily processed) refined carbohydrates, saturated fats, and sugar sweetened beverages that are cheaper, on a per-calorie basis, than fruits and vegetables, complex carbohydrates, and healthier fats.[7,29–35] The stress inherent to food insecurity works against dietary modification. In experiments and real-world longitudinal studies, stress leads to intake of more energy-dense but less healthful foods.[36–43] Food insecurity often occurs in those with low neighborhood access to healthy foods, and competing time demands that further incentivize convenience foods.[44,45] Food insecurity thus results in lower diet quality[7] and greater body weight.[29,46] Food insecurity increases hypoglycemia risk, regardless of HbA1c level, as patients often continue the same medication dose despite variable food intake.[47–50] Food insecurity leads to cost-related medication underuse (taking less medication than prescribed owing to cost), resulting in worse glycemic control.[11,24,51–57] Without addressing food insecurity, when present, glycemic control is unlikely to improve.

Rigor of Prior Research and Rationale for Current Study. As noted above, strong evidence demonstrates the associations between food insecurity, poor glycemic control, and other adverse T2DM outcomes. However, evidence that food insecurity interventions improve T2DM outcomes is lacking or not of sufficient rigor to guide clinical practice. To address this critical gap in knowledge, we propose a randomized clinical trial (RCT) of a medically tailored meal (MTM) intervention for adults with T2DM and food insecurity.

There are few randomized trials evaluating food insecurity interventions in individuals with T2DM. The key gap in rigor of past work is that there are very few RCTs for food insecurity interventions in individuals with T2DM (or other conditions). Owing to lack of RCT evidence, the optimal intervention approach is not known.[44,58,59] RCTs examining interventions for individuals with T2DM and lower socioeconomic status exist (e.g., peer-support, care management, or community health worker interventions).[60–66] However, these are not necessarily effective in a food insecure population, as over 50% of those with lower socioeconomic status are nevertheless food secure.[2] To our knowledge, aside from our pilot RCT[67], only one published RCT has evaluated a food insecurity intervention for T2DM.[68] This food pantry-based intervention, which provided boxes of healthy food, did not improve HbA1c, likely owing to other barriers to T2DM management (e.g., the time and skills needed to prepare appropriate meals). Our proposal explicitly addresses these additional barriers.

Food subsidies, the most common food insecurity intervention, have little evidence of impact on T2DM outcomes. Food subsidies, which offer cash or near-cash benefits to be spent on food, are the most common food insecurity intervention in the US.[35,44] Both SNAP (the Supplemental Nutrition Assistance Program, formerly the Food Stamp Program), and WIC (the Special Supplemental Program for Woman, Infants, and Children), the two largest food security programs in the US that include adults, are

food subsidy programs. RCTs of subsidies or incentives for healthy foods, including one from our group (NCT 03231592) have been conducted outside the context of T2DM care.[69,70] These have found that subsidies of $20-$40 per month can lead to improvements in diet quality of approximately 4 points on the Healthy Eating Index 2010 (HEI) scale.[69–71] HEI scores range from 0-100, with higher scores representing better diet quality; a 3 point difference is clinically meaningful.[30,71,72] One non-randomized study in T2DM found that a subsidy reduced HbA1c by 0.7%, but this study's rigor was limited by lack of randomization, pre/post design, and loss to follow-up.[73] Apart from food subsides, the other common food insecurity intervention is 'linkage' of patients to community resources (e.g., food pantries).[74,75] However, we found this approach did not improve HbA1c.[76]

Pilot RCT: From 6/2015 to 7/2017, we conducted a pilot RCT of MTM for individuals with T2DM and food insecurity in partnership with Community Servings (CS) (NCT02426138, PI: Berkowitz). The findings of this study, the first published RCT[67] of MTM for T2DM, support the rationale for the proposed work. This pilot was specifically designed to test proof of concept: whether meal delivery could improve diet quality. In this 24-week cross-over trial, participants received 12 weeks of MTM either immediately or after a 12-week delay (the 'on-meals' period), with the other 12 weeks serving as the control period ('off-meals'). The primary outcome was HEI score. Secondarily, we examined patient-reported outcomes. We also conducted semi-structured interviews (N=20), to better understand participant experience and to refine the intervention.

Pilot RCT results: Forty-four individuals enrolled and 2 withdrew, citing difficulty attending in-person study visits (N=42). Study retention was excellent: primary outcome assessment was completed for all participants. MTM strongly increased diet quality: mean 'on-meal' HEI = 71.3 (SD 7.5); mean 'off-meal' HEI 39.9 (SD 7.8); difference 31.4, 95%CI 26.7 to 36.1, p<0.0001. MTM reduced food insecurity (OR 0.44, p = 0.047) and hypoglycemia (OR 0.49, p = 0.03), and improved mental health-related quality of life (4 additional 'healthy days'[77] per month, p = 0.03). Participants reported reduced stress of managing T2DM and increased confidence in self-management, and used the meals as a model for subsequent healthy eating. Although the pilot study was not designed to detect changes in HbA1c, these changes suggest the intervention may be an effective way to do.

Additional Prior MTM Studies: Apart from the pilot RCT, several studies support the promise of MTM interventions. In two longitudinal cohort studies from our group, we found that MTM was associated with fewer inpatient admissions and lower healthcare spending, even when adjusting for potential confounders and accounting for the cost of the intervention.[78,79] Other non-randomized studies of MTM found lower healthcare costs,[80] reductions in diabetes distress, and improvements in diabetes management self-efficacy.[81]

Gaps in Rigor of Pilot RCT: As a pilot study, the key gaps in rigor of the pilot RCT are that it was neither large enough nor long enough to robustly evaluate the MTM intervention for its ability to reduce HbA1c.

Summary of background: This proposal is guided by an overarching scientific question: **"Is MTM efficacious for those with T2DM and food insecurity?"** MTM has a rising public profile that has outstripped its supporting evidence. Enthusiasm for MTM has led to calls for its use in the popular press.[82] MTM is part of Medicaid reform efforts in California, Massachusetts, and North Carolina.[83–85] Thus it is critical to understand the efficacy of MTM. If MTM works as intended, study findings will help catalyze widespread implementation. If it does not, or works only in limited contexts, then scare resources should be used to help disadvantaged populations in other ways. Thus, the results of this trial

will be highly informative, whatever the outcome.

2.3 RISK/BENEFIT ASSESSMENT

The potential risks to participants are few. The primary goal of the intervention is to improve food security and eat a healthful diet. By virtue of study eligibility criteria, participants are at higher underlying risk of hypoglycemia (owing to food insecurity), but this is likely to be reduced by participating in the study as both study arms provide resources to increase food security. In addition, the risks of phlebotomy, though uncommon, include pain, fainting, local bruising, and infection at the puncture site. We do not foresee significant psychological, financial, legal or other risks from participation in this research study.

2.3.1 KNOWN POTENTIAL BENEFITS

Cardiovascular disease (CVD) is the most common cause of mortality for patients with diabetes and rates of CVD are high for those with diabetes and food insecurity. There are high quality data suggesting that adhering to a healthful dietary pattern is associated with a substantial reduction in risk from heart disease and stroke. In addition, if the intervention improves glycemia, as expected, it will reduce the risk of microvascular and macrovascular complications.

2.3.2 ASSESSMENT OF POTENTIAL RISKS AND BENEFITS

As described above, the risk or taking part in this study is minimal, while the potential benefits are substantial. Overall, we believe the risks of participation are small compared to the potential benefits from participating in this study.

3 OBJECTIVES AND ENDPOINTS

OBJECTIVES	ENDPOINTS	JUSTIFICATION FOR ENDPOINTS	PUTATIVE MECHANISMS OF ACTION
Primary			
Primary Aim (efficacy): Hemoglobin A1c Hypothesis: We hypothesize that the MTM group will have 0.5% lower hemoglobin A1c at the end of treatment (6 months, primary analysis) compared with the food subsidy group.	HbA1c	Primary outcome.	Improvements via nutritional (diet quality, adiposity), compensatory (medication adherence, food-medication trade-offs) and psychological (self-efficacy,

OBJECTIVES	ENDPOINTS	JUSTIFICATION FOR ENDPOINTS	PUTATIVE MECHANISMS OF ACTION
			depressive symptoms, stress and distress) pathways
Secondary			
Secondary glycemic outcomes Hemoglobin A1c Hypothesis: We hypothesize that the MTM group will have 0.5% lower hemoglobin A1c 6 months after the end of treatment (12 months after start of study), compared with the food subsidy group.	HbA1c	Whether any intervention effect is durable after the end of intervention is important for understanding long-term health effects	As above
Secondary patient-reported outcomes We hypothesize that the MTM group will have less food insecurity, hypoglycemia, and diabetes distress, and greater diabetes self-efficacy, diet self-efficacy, and health-related quality of life, at the end of treatment (6 months) and 6 months after treatment ends (12 months), compared with the 'usual care + food subsidy' group.	See measures table for detailed list	These are important, relevant patient-reported outcomes.	As above
Secondary mediation outcomes Quantitative analyses will evaluate the 'how' by examining the mediating effect of changes in diet/weight, compensatory	Indicators of putative pathways (see measures table for detailed list)	These will help us understand why we observed the results we did	As above

OBJECTIVES	ENDPOINTS	JUSTIFICATION FOR ENDPOINTS	PUTATIVE MECHANISMS OF ACTION
strategies, and psychological factors.			
Secondary qualitative analyses To assess why, how, and under what circumstances MTM participants A) achieve improvements in T2DM outcomes and B) maintain these improvements after the intervention.	Qualitative interviews with participants (N= ~100) at 6 and 12 months will examine cognitive ('why?'), procedural ('how?'), and contextual ('under what circumstances?') dimensions of their response, if any, to intervention.	These will help identify potential mechanisms and moderators of treatment effects, such as race/ethnicity, and inform future modifications.	As above

Measures Table

Outcome/Measure	Pathway	Scale	Auxiliary scales, if any
Hemoglobin A1c	n/a	Continuous	Binary (> 9.0%), Binary (> 8.0%),
Food insecurity (Household Food Security Survey Module)[86]	n/a	Continuous	Ordinal (4-level), Ordinal (2-level)
Health-related quality of life[87–90] and domain scores (e.g., depressive symptoms)	n/a	Continuous	t-scores for each domain, and PROMIS-Preference (PROPr) score
Diabetes Distress (PAID 11)[91]	n/a	Continuous	
Self-reported hypoglycemia and severe hypoglycemia[92]	n/a	Categorical	
Systolic blood pressure	n/a	Continuous	Binary (< 140 mm Hg)
Diastolic blood pressure	n/a	Continuous	Binary (< 90 mm Hg)
Body weight	Nutritional	Continuous	% change, Binary (≥5% change)
Diet quality[93,94]	Nutritional	Continuous	Subscales
Housing instability[95]	Compensatory	Categorical	
Transportation barriers[95]	Compensatory	Categorical	
Delay care[95]	Compensatory	Categorical	
Financial strain[95]	Compensatory	Categorical	
Medication adherence (1-item SDSCA[96])	Compensatory	Continuous	
Medication adherence (7-item ARMS-D[97])	Compensatory	Continuous	
Cost-related medication underuse[51,54,98,99] (NHIS items)	Compensatory	Categorical	
Food/medication trade-offs[54]	Compensatory	Categorical	
Perceived stress[100]	Psychological	Continuous	
Diabetes self-efficacy[101]	Psychological	Continuous	
Diet self-efficacy[102]	Psychological	Continuous	

4 STUDY DESIGN

4.1 OVERALL DESIGN

The proposed single-site, open-label, Phase II, community-based randomized clinical explanatory trial will be used to evaluate the efficacy of a medically tailored meal (MTM) interventions for adults with food insecurity and T2DM. While the efficacy of MTM needs to be established, given the nature of the population served by MTM, it is important to design the study to enhance the external validity / generalizability of the findings as much as possible. We will enroll N=200 participants recruited from the referral list of Community Servings, a Boston-based MTM provider, and assign them by 1:1 randomization to the two treatment regimens: MTM intervention (N=100), versus the control regimen of usual care + monthly food subsidy (N=100). MTM intervention will consist of 6 months of weekly delivery of 10 ready-to-eat meals (approximately half an individual's weekly food intake), along with telephone-based lifestyle intervention that prepares participants for the post-treatment period. The usual care + food subsidy recipients will receive usual care along with 6 months of a $40/month food subsidy.

4.2 SCIENTIFIC RATIONALE FOR STUDY DESIGN

Food insecurity worsens T2DM outcomes through 3 key pathways: nutritional, compensatory, and psychological. Our understanding of the expected effect of MTM on T2DM outcomes is based on social cognitive theory[103,104] and our past work understanding how food insecurity worsens health in T2DM.[34,44,59] Based on this empirical evidence, there are 3 key pathways whereby food insecurity serves to worsen T2DM outcomes. First, in the *nutritional pathway*, food insecurity is associated with lower diet quality, which is in turn associated with greater HbA1c. Food insecurity incentivizes more affordable foods that both directly raise serum glucose (e.g., high consumption of refined carbohydrates and low fiber) and may lead to greater insulin resistance.[105,106] Second, via a *compensatory pathway*, behavioral strategies necessary to cope with the immediate problem of food insecurity can inadvertently undermine T2DM management. For example, financial resources that might otherwise have been used for medications or diabetes care supplies are diverted to meet dietary needs.[51,107] Third, through the *psychological pathway*, the state of food insecurity, in which meeting basic needs is outside an individual's control, undermines self-efficacy, and increases depressive symptoms and diabetes distress.[24,108–110]

The field of food insecurity interventions for T2DM is in a state of scientific equipoise. Food insecurity is clearly associated with poor T2DM outcomes, but the best way to intervene is unknown. The next logical scientific step is to test the ability of MTM to improve T2DM outcomes in randomized clinical trial that is larger-scale and longer-term than our prior pilot study. Doing so will advance the field by helping to answer the following pressing scientific questions:

1) Can a longer duration MTM intervention produce changes in HbA1c and patient-reported outcomes during treatment (Aim 1a)? This project will assess the effect of MTM on the primary

outcome of HbA1c at the end of treatment (6 months). Further, we will determine whether MTM improves patient-reported outcomes, including hypoglycemia, health-related quality of life, depressive symptoms, self-efficacy, and diabetes distress, at the end of treatment (6 months).

2) Can an MTM intervention produce sustained improvements in HbA1c and patient-reported outcomes post-treatment (Aim 1b)? All MTM intervention combines meal provision with education. The lifestyle component of our MTM intervention combines education and skill-building for sustained behavior change, and the delivered meals serve as a model for healthy eating.[111–113] Thus MTM may improve T2DM outcomes even after the intervention, which we will assess at 12 months (6 months after meal delivery ends).

3) Why, how, and under what circumstances does an MTM intervention improve outcomes among individuals with T2DM and food insecurity (Aim 2)? Although the mechanisms underlying the relationship between food insecurity and poor T2DM outcomes are well understood, less is known about how MTM may improve outcomes. Differences in demographic characteristics and social circumstances may all influence MTM response. This underscores the need to better understand mechanisms of action and contextual factors that may influence on-treatment and post-treatment response. Aim 2 will use quantitative analyses to assess pre-specified putative mediators (see C.13 & C.15.2), and qualitative interviews to examine cognitive ('why?'), procedural ('how?'), and contextual ('under what circumstances?') dimensions of participants' response to MTM (see C.15.1.) to uncover novel mechanisms for A) on-treatment and B) post-treatment effects. These mechanistic findings will aid in understanding the efficacy results and inform future modifications of MTM.

4.3 JUSTIFICATION FOR INTERVENTION

MTM addresses the 3 key pathways by which food insecurity worsens T2DM outcomes. MTM provides diabetes-appropriate food, allows resources previously spent on food to be re-allocated to T2DM management, and stabilizes the uncertain circumstances of food insecurity. MTM is likely to have both on-treatment and post-treatment effects. On treatment, MTM both provides healthy meals and T2DM self-management skill-building via lifestyle intervention. This is likely to result in changes to mediators of T2DM outcomes—improved diet quality, reduced trade-offs between food and medication, and improvement in psychological factors. As meal delivery ends, the motivation, problem-solving, and self-regulation skills developed during MTM become more salient. The meals received are a model for how to construct and portion meals, and expose participants to new foods. Further, the lifestyle intervention component of MTM helps individuals avoid maladaptive compensatory behaviors, such as cost-related medication underuse, and increase diet and diabetes self-efficacy, promoting resilience. In summary, MTM is highly likely to be efficacious because it removes as many barriers to healthy eating as possible, off-loads household budgets to free-up scarce resources, models T2DM-appropriate meals, and leverages the period of meal delivery to facilitate behavioral change and skill-building.

MTM is likely to be more efficacious than other food insecurity interventions because MTM addresses more barriers to T2DM management. Food subsidies require individuals to purchase and prepare healthy foods for themselves and may not sufficiently change food intake to affect clinical outcomes.[114] Limitations relating to knowledge and experience of appropriate foods, neighborhood food access,

motivation, and time-constraints[115] all may serve to lessen the effect of subsidies. Because our MTM intervention is designed to provide food, education, and behavioral skills, it is likely to overcome these additional barriers.

4.4 END-OF-STUDY DEFINITION

The end of study is defined as completing the 12 month follow-up visit.

5 TARGET POPULATION AND STUDY SAMPLE

5.1 INCLUSION CRITERIA

Inclusion Criteria
- Male or female
- Diagnosis of Type 2 Diabetes Mellitus
- Hemoglobin A1c ≥ 7.0% and ≤ 12.0% in last 12 months
- Experiencing food insecurity as indicated by 2-item Hunger Vital Sign
- English speaking
- Aged ≥18 years
- BMI ≥25 kg/m2 (≥ 23 kg/m2 if self-report Asian ancestry)
- No plans to move from the area for at least 1 year
- Free living to the extent that participant has control over dietary intake
- Willing and able to provide written informed consent and participate in all study activities

5.2 EXCLUSION CRITERIA

- Participant in diabetes, nutrition, or weight research intervention in last 12 months
- Another family member or household member is a study participant. Only one member of each household may take part in this study.
- Considering bariatric surgery in the next year or prior bariatric surgery in the past 2 years
- Lack of safe, stable residence and ability to store meals
- Lack of telephone
- Pregnancy/breastfeeding or intended pregnancy in the next year
- History of malignancy, other than non-melanoma skin cancer, unless surgically or medically cured > 5 years ago or in remission. Patients with localized prostate and breast cancer diagnosed during the course of routine screening will not be excluded.
- Advanced kidney disease (estimated creatinine clearance < 30 ml/min)
- Known drug or alcohol misuse in the past 2 years
- Known psychosis or major psychiatric illness that prevents participation with study activities
- Intermittent use of medications (e.g., oral or intravenous glucocorticoids) that are likely to affect blood sugar

5.3 LIFESTYLE CONSIDERATIONS

During this study, participants may be asked to:

- Follow the lifestyle recommendations (e.g., for diet and physical activity).
- If they are unable to make the recommended changes, they will continue in the study. Specifically, participants will not be excluded if they fail to achieve lifestyle goals.

5.4 SCREEN FAILURES

As noted in Sections 5.1 and 5.2, potential participants will be carefully screened for interest and eligibility before the enrollment. We anticipate the number of "screen failures" at the enrollment visit will be small.

5.5 STRATEGIES FOR RECRUITMENT AND RETENTION

Recruitment plan

Availability of potential participants: Study participants will be recruited from Community Servings' (CS) list of referred adults.

Anticipated yield from recruitment and screening efforts: Our goal is to enroll, on average, two individuals per week over a 24-month recruitment period. Factors that will enhance recruitment are that CS' referral list contains individuals who have already indicated that would like to enroll in CS' services, and that the study imposes few additional obligations for the participants. Further, because study participation guarantees receiving either the MTM intervention or the food subsidy, many individuals who would not otherwise receive a food insecurity intervention would be able to do so as part of the study. Finally, we will convene an advisory board of current CS clients who will provide advice on maximizing recruitment. We will incorporate this advice into our recruitment strategies. Taken together, we estimate that approximately half of eligible participants will wish to participate in the study, or a yield of 50%. However, a yield as low as 28% would be sufficient to meet recruitment targets in the planned 24-month time frame.

Back-up recruitment strategy: If recruitment lags, we will solicit referrals to CS from local diabetes centers and primary care offices to augment the pool of eligible participants, and leverage CS' existing relationships with free clinics and local health departments.

Inviting identified candidates to participate: We will contact potential participants by phone to assess interest and eligibility for participation. If interested and eligible, an enrollment visit will be scheduled.

Also, the study consent form will be mailed or emailed to the potential participant before the enrollment visit, to allow time for review.

Enrollment visit and randomization: At the enrollment visit, all questions will be answered about the study and those willing to take part will sign the consent form. Upon completion of baseline measures the participant will be ready for randomization. The research assistant will use a secure web-portal to ascertain randomization assignment.

Retention plan

Several methods will be used to promote participant retention. We also note that though we will promote participation in a timely fashion as much as possible, we will also respect study participants' desire to withdraw from the study at any time.

Rapport with study staff: We found in our pilot study that establishing rapport with study staff greatly facilitates successful completion of the study. Our study staff training will emphasize the importance of this.

Multi-modality reminders: To enhance retention of participants, at enrollment, we will ask for home and cell numbers, if it is OK to send text messages, and if it is OK to contact by email. We will also ask participants to provide contact information for an alternate contact (friend or family member) in case we are unable to reach them directly. In attempting to schedule a follow-up contact, we will allow up to 5 attempts to reach a participant over a 2-week period and up to 3 attempts to reach the alternative contact. Our goal is to contact participants who are late for study activities early and often in support of their participation. We will use multiple methods of reminding participants about study appointments, including written reminder slips, reminder phone calls to their home and/or mobile phone, and/or reminder text messages to their mobile phone, emails, and mailings. We successfully used each of these strategies in our pilot RCT.

Financial incentives for outcome assessments: We will remunerate participants in both arms $20 for completing a 3 month check-in, $20 for completing a 6 month outcome assessment and $50 for completing a 12 month outcome assessment. This will help to minimize missing outcome data.

Location of intervention / assessments: The home-based setting of our RCT places the intervention as well as assessment visits in the most convenient location. We will also conduct as many of our assessments as possible by telephone, to maximize convenience for the participant. This will enhance the ability of participants to remain engaged in the study for the planned duration of follow-up.

Community-engaged planning: In addition to the above strategies, we will seek input from pre-trial focus groups on other ways to enhance participation and retention in the proposed trial.

Possible competition from other trials, secular trends, and co-interventions: We do not anticipate major competition from other trials of food insecurity interventions. Community Servings, as our partner in this trial, has no plans to participate in other research interventions that would complete with this trial. It is unlikely that secular trends will have a major impact on food insecurity, but if they do, they should have similar impacts across our study arms. Similarly, co-interventions such as diabetes medications or use of government food security program like SNAP would likely be equally distributed across study groups. If not, we could adjust for any covariate showing imbalance across groups in our analyses.

Expected Challenges and Contingency Plans. If recruitment lags, we will solicit referrals to CS from local diabetes centers and primary care offices to augment the pool of eligible participants, and leverage CS' existing relationships with free clinics and local health departments. Regarding retention, a key challenge that can occur in this population is frequent moves. Because CS provides a home delivered meal service, those referred are generally less likely to move than the background population, but this could still occur. As long as individuals move within the CS service area (which encompasses the state of Massachusetts and part of Rhode Island), study activities can continue. Should individuals move outside the service area, they would be unable to continue participation in the study if there move was during the initial 6 months (active intervention period). If the individuals were to move after the initial 6 months, we would attempt to complete study follow-up activities as planned (especially as many activities can be completed by telephone). We did not encounter issues with participants changing phone numbers during the pilot study, but this could occur. Rapport with study staff is key to minimizing loss to follow-up, and will be emphasized by our training. We will record, to the best of our ability, reasons for discontinuation, so that these can be adjusted for as part of our missing data strategy.

6 STUDY INTERVENTION(S) OR EXPERIMENTAL MANIPULATION(S)

6.1 STUDY INTERVENTION(S) OR EXPERIMENTAL MANIPULATION(S) ADMINISTRATION

6.1.1 STUDY INTERVENTION OR EXPERIMENTAL MANIPULATION DESCRIPTION

Medically Tailored Meals: The MTM intervention consists of weekly home meal delivery; an explanation of the medical tailoring of the meals; and a 6-session telephone lifestyle intervention change program designed to complement the period of meal delivery and prepare for the period after meal delivery with behavioral and skill-building approaches to sustain the benefit of the intervention.
Meal Delivery: During the 6 months of meal provision, 10 medically tailored meals (5 lunches and 5 dinners) will be delivered once weekly. Meal composition will follow ADA guidelines, and, as per ADA guidelines, an initial meeting with an RDN at enrollment will help individualize the meal planning to the specific "preferences, goals, and needs of the individual,"[116] keeping in mind any co-morbidities. To help individualize these meals, Community Servings can provide over 15 meal 'tracks', with up to 3 combinations of tracks being possible for a given individual. In this study, all individuals will have T2DM, and so will receive a diabetes-appropriate diet, but may also receive tailoring for other co-morbidities. For example, they may be on warfarin anti-coagulation, requiring a consistent level of vitamin K in the

diet, which can be accommodated. Calorie targets are set based on current body weight and goals. With regards to the diabetes-appropriate diet, the default plans follows the DASH pattern, as this has been associated with a clinically significant decrease in HbA1c in an RCT,[117] but other evidence-based patterns such as Mediterranean-style or plant-based can be accommodated.[116] Goal sodium is < 2300 mg daily.[116] The 10 meals represent approximately half of an individual's weekly food intake (3 meals per day x 7 days = 21 meals weekly), which is in keeping with the goal of supplementing, rather than fully replacing, an individual's food intake. Supplementation is a guiding principal for most nutrition assistance programs (e.g., SNAP or WIC). Meals are refrigerated or frozen and can be heated and served with no special cooking equipment and minimal preparation time. To ensure cultural acceptability, CS solicits family recipes from clients (adapting them to meet nutritional goals). In the pilot program, participants were very enthusiastic about the taste and cultural acceptability of the meals, and many participants wanted to keep receiving the meals after the end of the study. Because the meals are tailored to the specific medical needs of the individual, participants are asked not to share meals with other household members. While it may seem unlikely that individuals facing food insecurity would not share food, particularly with their children, what we have seen happen in practice is that food is not typically shared. This is chiefly because the money that would otherwise have been spent on food for the participant can be re-allocated to food for other household members. Also, the medical tailoring of the food likely plays a role in this as well, with other household members recognizing the role of the meals in the individual's treatment plan.

1) Lifestyle Intervention: ***Theoretical model and rationale.*** In keeping with our conceptual model, the intervention focuses on three key areas relevant to food insecurity and T2DM: nutrition and diet quality, compensatory behaviors, and psychological factors. The proposed lifestyle change intervention, like the Look AHEAD lifestyle intervention, is based on the Social Cognitive Theory (SCT) of behavior change. SCT outlines basic principles that inform most behavior change interventions, which in turn, underlie the lifestyle change skill building strategies that will be used, including: (1) acquisition of knowledge; (2) self-monitoring; (3) realistic, incremental goal-setting; (4) problem solving; (5) stimulus control; (6) cognitive restructuring; and (7) providing support for change. These strategies help build self-efficacy and flexible dietary restraint skills, i.e., continuing to manage a healthy eating pattern in the face of stressors like food insecurity.[117,118118,119118,119118,119118,119] The 6-session lifestyle intervention is targeted to the needs of those with T2DM experiencing food insecurity, with an emphasis on improving glycemic control. This invention will have a primary focus on diet quality, not weight loss. Our rationale for this is 1) in our prior work with food insecure populations, participants noted that the intensity of an effective weight loss intervention (e.g., 16-19 sessions in 6 months) would be difficult to adhere to in the context of competing demands, and 2) that such intensity would be difficult for most MTM agencies to provide (and thus not generalizable). However, participants who do wish to lose weight can select a weight loss option at their initial meeting with an RDN for individualized goal setting. Besides improvements in diet quality, other pathways to glucose lowering that the intervention addresses are compensatory behaviors and psychological factors, as detailed in our conceptual model. Therefore, we anticipate that the intervention will result in lower glycemia with or without weight loss. The intervention we propose aims to facilitate real-world implementation and minimize barriers to participation. The intervention materials are culturally sensitive and target a Flesch-Kincaid grade level score <6. The materials build skills that participants apply to their food preferences.

Planned Contacts

	Number of Planned Contacts	Contact Duration	Timing	Estimated Total Contact Time
Lifestyle Intervention Sessions	6	Phone calls = 30-45 min.	After randomization and monthly thereafter	3-4.5 hours
Minor Check-ins	6	Phone follow-up = 15 min.	1-2 weeks after Lifestyle Intervention Session	1.5 hours
TOTAL	**12**	**--**	**--**	**4.5-6 hours**

Session Content and Delivery: Each session will include components that address nutrition knowledge and dietary behaviors, along with cognitive and behavioral content shown to be important to promoting short- and long-term lifestyle change. Throughout, nutrition and lifestyle counseling will focus on self-monitoring, self-efficacy building, problem solving, and goal setting with action planning, and will be tailored to the circumstances experienced by the participant. Program goals for dietary pattern behaviors will guide the feedback provided to participants during these monthly program sessions as well as the shorter check-in contacts.

The following content is addressed during the 6 lifestyle intervention sessions:

Session #	Content
1	• Welcome • Healthy Eating • Healthy Foods Guide • Healthy Eating on a Budget • Move those Muscles
2	• Working with What's Around You • How to Read a Nutrition Label • Better For You Products
3	• Ten Ways to Control Your Hunger • Drinks, Desserts, Snacks, Eating Out and Salt
4	• Mid-Point Progress Review • Better For You Fast Food • Ways to Improve Restaurant Choices • Problem-solving
5	• Build Better Light Meals • The Slippery Slope of Lifestyle Change

6	• Ways to Stay Motivated • Making Time for the Important Things

At the first session, an overview of the program is given and then the specific content for Session 1 is presented. Individualized calorie targets are set based on stated goals, with option to maintain weight with focus on diet quality or focus on diet quality and weight loss. Participants are instructed on how to complete a food and activity diary and record self-monitoring of weight and results of any self-blood glucose monitoring. At the end of the session, the participant and counselor discuss an individually tailored action plan to help guide the participant's eating activity behaviors for the next month (or until the next counseling session). Finally, the counselor and participant identify 2 achievable goals (participants could opt to choose 1) to work on before the next session and document them on a goal sheet.

Subsequent sessions open with a check-in to review self-monitoring of weight, eating, activity and blood glucose; medication adherence and progress toward behavioral goals. Then, the counselor and participant engage in an interactive discussion of session content, review any practice worksheets to be completed and agree upon achievable goals for the next session.

The lifestyle intervention sessions, above, include behavioral components and a set of cognitive skills that have been shown to be important for maintenance of behavior change after interventions end.[119,120] These skills build self-efficacy and increase intrinsic motivation to engage in lifestyle and self-regulation behaviors. The table below highlights these skills by a brief summary of skills and examples of activities/discussions that will be integrated into the monthly sessions.

Longer-term dietary change is best achieved by addressing the *challenges to self-regulation* of diet. The underlying premise is that the intrinsic desires to consume palatable, high-calorie foods and the experience of food insecurity, which incentivizes easily accessible unhealthy foods, make it hard for individuals to maintain diet quality. Both external (e.g., sight of food) and internal cues (e.g., cravings, feelings, and thoughts) may promote eating of less healthful foods, especially in those experiencing the additional stressor of food insecurity.

For this proposed explanatory trial with a primary focus on dietary pattern and resiliency to food insecurity, we do not propose a high intensity physical activity intervention component. Instead, we will periodically mention physical activity recommendations consistent with national guidelines.

Content of Minor Check-ins

Timing	Content
1-2 weeks after Session; ~15 minutes per contact	• Review self-monitoring (diet, physical activity, medication adherence, weight and blood glucose) • Discuss successes and challenges with achieving goals set by participant.

	• Engage in problem-solving to address behavioral challenges • Provide feedback on progress relative to goals • Set goals and develop action plans for the following period

Safety Monitoring: Because the intervention simply provides healthy food to those with T2DM and food insecurity, we anticipate few adverse events from the intervention itself. However, the study sample is likely to face some risks due to their overall circumstances. Key risks we anticipate are low blood sugar and the possible need for medication adjustment. Though we saw reductions in hypoglycemia during the pilot trial[67], those with food insecurity and T2DM are at high risk for this adverse event. During the baseline visit, study staff will discuss signs and symptoms of hypoglycemia, and remind participants to develop a treatment plan for this with their treating physician. Medical management of hypoglycemia will be under the purview of the participant's care provider. Referral to CS requires a care provider certification so all study participants will have a care provider. Also, participants in this study will reside in Massachusetts where health insurance coverage is nearly 100%, so lack of health insurance is not commonly a barrier to healthcare access. However, study staff will record hypoglycemic events and communicate them to the treating physician, to adjust treatment plans as needed. These calls will occur monthly, as part of the 'check-in' call 1-2 weeks after the lifestyle intervention session

Usual Care + Food Subsidy: As a comparison group, those not randomized to receive the MTM intervention will receive usual care plus a food subsidy for 6 months.
Rationale for the usual care + food subsidy comparison group: Though food insecurity often goes untreated in clinical care, not offering treatment for food insecurity in individuals with T2DM in a research setting poses unacceptably high risks to participants. Therefore, we propose to augment usual care by offering a $40/month food subsidy.

Usual Care: In the context of this trial, usual care will consist of care provided by the participant's typical medical team and will not be managed by study personnel. We will ask participants to perform glucose self-monitoring in accordance with their diabetes care plan. To ensure participants are aware of T2DM care recommendations, we will provide participants with lay language educational materials about T2DM goals (taken from educational materials made available by the Centers for Disease Control and Prevention [*https://www.cdc.gov/diabetes/basics/type2.html*]). Further, we will provide individuals with information regarding recommendations for physical activity and Medical Nutrition Therapy, with instructions that they discuss this with their diabetes care provider if interested.

Food Subsidy Logistics: Food subsidies will consist of $40/month gift cards to local grocery stores. Further, participants will be provided with guidance related to using the funds to purchase foods that improve glycemic control, including shopping tips and healthy recipes. To address issues of high contact between study staff and MTM participants, study staff will use each 30-day 'reload' of the food subsidy as a time point to contact food subsidy participants, troubleshoot issues with using the subsidies,

remind the participant of the healthy food guidance that underpins the subsidy, and collect information about food purchases. These check-ins help ensure more balanced contact with study staff between study arms.

Safety Monitoring: Safety monitoring will follow the same protocol as for the MTM group and occur on the same schedule (1-2 weeks after the monthly phone call) to balance staff contact between study arms.

6.1.2 ADMINISTRATION AND/OR DOSING

Please see description above

6.2 FIDELITY

6.2.1 INTERVENTIONIST TRAINING AND TRACKING

Protocols for Intervention Delivery

Protocols to guide the delivery of the proposed lifestyle intervention will be adapted from the pilot RCT and prior lifestyle interventions trials conducted by the research team. Key protocols that guide the delivery of the study intervention will address intervention staff training, feedback and monitoring, and participant care management. Intervention delivery staff will receive training that includes: intervention and study design, T2DM treatment guidelines, nutrition and physical activity recommendations, adult behavior change principles and adult learning, motivational interviewing principles, counseling strategies specific to intervention delivery, review of the intervention leader's guide and participant materials. Training will include opportunities to practice counseling skills. During the implementation of the intervention, the registered dietitian nutritionists (RDN) who will deliver the intervention will meet regularly with Ms. Delahanty and other study staff to review delivery of intervention contacts, share successful strategies, and problem-solve to address challenges.

Monitoring for Quality and Consistency of the Intervention

Fidelity of intervention delivery will be assessed using content and process checklists. A portion of intervention sessions (up to 20%) will be either monitored in real-time by Ms. Delahanty or audio recorded with asynchronous review by Ms. Delahanty. With each monitored intervention session, RDNs will receive timely feedback and supplemental training based on observed issues, if any. These brief trainings will be included in regularly scheduled meetings with RDNs.

6.3 MEASURES TO MINIMIZE BIAS: RANDOMIZATION AND BLINDING

Randomization

At the end of the enrollment visit, after all data are collected, enrolled participants will be randomly assigned to MTM intervention (N=100) or monthly food subsidy (N=100) in a 1:1 ratio. To increase

precision of effect estimates, randomization will be stratified by gender, race/ethnicity (non-Hispanic white vs. not), and whether HbA1c at time of referral is above or at/below the median (8.6%) of CS' eligible clients. Randomization tables will be prepared by a biostatistician not affiliated with the study using a permuted block algorithm with blocks of size 2 and 4 in unpredictable order. REDCap will be used to conceal the treatment assignments from all study personnel until the moment a newly enrolled participant is randomized. Randomization programs will be written by the study statistician, but will be implemented by a statistician not otherwise involved with the study, using their own selected seed, and the generated allocation sequence will be stored electronically in a password-protected file.

Unblinding Plan
As an open-label study, participants, providers, and many study staff will know to which group individual participants have been randomized. However, efforts will be made to keep assay evaluations, outcome assessors, and all data analysts blinded to treatment regimen assignments during the course of the trial. Any key decisions regarding study outcomes, methods, and changes to the protocol and statistical analysis plans will be made in a masked review of the data.

Statistical computations will be initially programmed using arbitrary treatment assignments. Unmasking with respect to the true randomization groups will only be done for the final interpretation of the results.

6.4 STUDY INTERVENTION/EXPERIMENTAL MANIPULATION ADHERENCE

- Please see above

6.5 CONCOMITANT THERAPY

Participation in this study does not preclude use of any other diabetes medications, or any other food insecurity interventions such as food pantries or the Supplemental Nutrition Assistance Program.

Bariatric Surgery

At enrollment, we assess interest in bariatric surgery and if a potential participant is contemplating this within the next year, he/she will not be enrolled. That said, a patient may elect to have bariatric surgery at any time during his/her participation in this study. If a participant has bariatric surgery, he/she will be withdrawn from the study at that time. Data collected prior to withdrawal will be included in the analysis.

6.5.1 RESCUE THERAPY

No rescue therapy outside of the study protocol is planned.

7 STUDY INTERVENTION/EXPERIMENTAL MANIPULATION DISCONTINUATION AND PARTICIPANT DISCONTINUATION/WITHDRAWAL

7.1 DISCONTINUATION OF STUDY INTERVENTION/EXPERIMENTAL MANIPULATION

The study intervention will not be discontinued unless the participant withdraws or is withdrawn as outlined in the above section.

If a participant withdraws or is withdrawn, no attempts will be made to collect follow-up measures.

If a participant requests to withdraw from the intervention but is willing to continue the measurement component of the study, he/she will be invited to continue with follow-up measures.

7.2 PARTICIPANT DISCONTINUATION/WITHDRAWAL FROM THE STUDY

Participants are free to withdraw from participation in the study at any time upon request.

Participants will be withdrawn according to protocol for the following reasons:

- New diagnosis of condition on study entry exclusion criteria list (type 1 diabetes and cancer, as defined on the list).
- Other medical diagnosis typically associated with significant weight loss (5% or more of body weight) that could affect the safety of continuing with the study. In addition to malignancy, examples include nonmalignant gastrointestinal disease, endocrinopathies, infectious disease, rheumatologic diseases, etc. Should this occur, we will seek adjudicate from medical colleagues not associated with the study and blinded to the participants study group. Any decisions of this nature will be documented in writing and retained as part of the study record.
- New diagnosis of cognitive decline confirmed by PCP as sufficient to interfere with insight related to food selection.
- New diagnosis of medical condition such that participation in a behavioral weight loss intervention is no longer appropriate.
- Permanent change in living situation (e.g. from home to nursing home) such that the participant has limited food options.

Pregnancy: Study participants who become pregnant will be withdrawn from the study as gestational diabetes is a separate condition, not under study here. That said, we do not anticipate any harm from study participation if a participant inadvertently continues with the study if she does not know she is pregnant.

7.3 LOST TO FOLLOW-UP

A participant will be considered lost to follow-up if we are unable to contact him/her according to this algorithm.

- Telephone contact:
  - We will call up to 5 times over a 2 week period. We will attempt 2 calls during regular work hours (8:30 to 5:30 on weekdays) and 3 after hours, including Mon-Thurs evening between 5:31 and 9:00 pm and on Saturday calling between 8:30 and 5:30 pm.
  - We will leave up to 2 phone messages.
  - If no contact is made over this initial 2 week period as outlined above, we will undertake more limited contact attempts for an additional 2 weeks including up to 2 phone calls per week, 1 during business and the other during evening hours or on the weekend.
- Email: If a participant gives us permission to contact by email, we will send up to 4 weekly emails.
- Text: If a participant gives us permission to contact by text, we will send up to 4 weekly texts.
- Alternative contact: During enrollment, we will collect information on an alternative contact that we are given permission to contact. We will attempt to contact this individual up to 3 times a week for 4 weeks.

If we are unable to contact a participant as outlined above, a letter will be sent to the provided address outlining our attempts to contact the participant and inviting the participant to contact us if he/she still wishes to participate in this study.

A participant who is lost to follow-up can still contribute electronic health record data to outcome datasets.

8 STUDY ASSESSMENTS AND PROCEDURES

8.1 ENDPOINT AND OTHER NON-SAFETY ASSESSMENTS

8.1.1: Screening, eligibility assessment, and consent

- As above, potentially eligible participants will be drawn from the list of those referred to Community Servings. Eligibility will be confirmed during an eligibility phone call. Those who wish to participate will provide informed consent, at which time baseline collection of study measures will occur (prior to randomization).

8.1.2: Overview of study measures

Study data will be collected during study sessions at baseline, 6, and 12. Process data for the MTM group will be collected during lifestyle counseling and check-in calls. Process data for the food subsidy group will be collected during the check-in calls. A subset of participants will be invited to complete semi-structured interviews after their 6 month and 12 month assessments. This is so that the interviews will

not influence data collected for those assessments. Owing to the potential for literacy limitations in the study sample, all data will be collected with the assistance of study staff (e.g., by reading questions aloud) and recorded electronically in REDCap. Study measures are described in Section 3, above. Where possible, measures are taken from the NIH-recommended PHENX Toolkit.

8.1.3: Detailed protocol for data collection

Note: Research staff will be trained and periodically retrained on data collection protocols. Baseline training will include detailed review of and practice with protocols under the direction of the study PI. Retraining will occur at quarterly intervals.

8.1.3.1: Weight and height protocols

- Scales tested for accuracy the first week of each month during their use in this study.

Seca 874/876 Scale Tolerance Testing

The scale's 6 AA batteries should be replaced at least once a year.
It is **imperative** that tolerance weights be handled carefully (no slamming around or banging against one another) to maintain their accuracy.
Before starting the test set the **ON-OFF** switch to **ON.** Press the **start** key with no load on the scale. If necessary switch the weight display from **kg** to **lbs** by pressing the **2 in 1** key for approximately 3 seconds.

1) Place a 50 lb wt in the center of the scale and record weight. (~50 lbs) Remove the 50 lb wt. Press **start** key.
2) Carefully stack two 50 lb wts on top of one another in the center of the scale. The first weight must be gently vibrated until the second is in place to prevent the scale from prematurely locking in and giving an erroneous reading. Record the weight reading (~ 100 lbs). Press the **2 in 1** key.
3) Place a third 50 lb wt on the scale and record weight. (~50 lbs The sum of this weight and the weight observed in step 2, ~ 100 lbs, is the observed reading for the expected weight of 150 lbs) **Remove <u>the top 50 lb wt. only</u>**.
4) Place two 50 lb wts on top of the two remaining weights vibrating the stack as before. Record the weight reading (~ 100 lbs The sum of this weight and the weight observed in step 2, ~ 100 lbs, is the observed reading for the expected weight of 200 lbs). Remove all four weights. Press **start** key.
5) Carefully, but quickly, stack five 50 lb wts on top of one another in the center of the scale, gently but firmly vibrating the stack until the fifth weight is added. Record the weight reading (~ 250 lbs). Press the **2 in 1** key.
6) Place a sixth 50 lb wt on the scale and record weight. (~50 lbs The sum of this weight and the weight observed in step 6, ~ 250 lbs, is the observed reading for the expected weight of 300 lbs) **Remove <u>the top 50 lb wt. only</u>**.
7) Place two 50 lb wts on top of the five remaining weights vibrating the stack as before. Record the weight reading (~ 100 lbs The sum of this weight and the weight observed in step 6, ~ 250 lbs, is the observed reading for the expected weight of 350 lbs). Remove all weights.

8) Set **ON-OFF** switch to **OFF.**

During the testing if any observed reading falls outside the tolerance range repeat the testing steps to verify the reading. No need to continue testing the scale if a weight is out of range and verified. Scales failing the tolerance testing should no longer be used.

lbs	**0.40% Tolerance Limit**	**Range**
50	0.2	49.8 - 50.2
100	0.4	99.6 - 100.4
150	0.6	149.4 – 150.6
200	0.8	199.2 – 200.8
250	1	249.0 – 251.0
300	1.2	298.8 – 301.2
350	1.4	348.6 – 351.4

Scale S/N _____

Date _____

Expected Weight (lbs.)	**Observed Weight (lbs.)**	**Difference (lbs.)**	**0.40% Tolerance Limit (lbs.)**
50			0.2
100			0.4
(50)			
150			0.6
(100)			
200			0.8
250			1
(50)			
300			1.2
(100)			

WEIGHT MEASUREMENT PROTOCOL
(last updated 3/5/21)

Prior to the appointment the participant is instructed to wear light clothing and informed that the weight measurements will be made with shoes removed.

Materials
- Seca model 874 or 876 electronic personal scale

Preparation
- Place the scale on a hard level surface.
- Turn the scale ON with the ON-OFF switch. Press the start key with no load on the scale. The scale is automatically set to zero and ready for use.
- Once the digital display reads zero **verify that "kg" unit is showing**. If "lbs" appears in the display press the 2 in 1 key for approximately 3 seconds. "kg" should then show in the display.

Procedure
Participant
- With shoes removed, scale displaying zero; step onto the middle of the scale.
- Stand on both feet, knees extended (straight) with arms by sides.

Research Staff
- Once the reading in the digital display is stable, record the first weight on the Weight Measurement Form in kg, **recording the tenths of kg** (i.e. 81.4 kg).
- Have the participant step off the scale.
- Have the participant step back onto the scale once zero is displayed. **Note:** You may need to restart the scale by pressing the start key to obtain the zero and kg display before instructing the participant to step back onto the scale.
- Once the reading in the digital display is stable, record the second weight on the form.
- Record the difference between the first and second readings.
- If the **difference is greater than 0.5 kg repeat** and record a third weight.
- When finished weighing participants turn the scale OFF with ON-OFF switch.
- Weight will be determined from the average of all recorded observations.

Note
- A stabilized reading in the scale display is needed for recording. The participant needs to stand very still while breathing naturally. If moving too much the weight will not stabilize in the display.

Scale Accuracy Testing

- Staff should weigh themselves or an object with a known weight whenever the scale is set up for participant measurement, to ensure scale is providing reasonable values.

Scale Care

- Treat the scale as a laptop. Do not bump against other objects or expose to extreme temperatures.
- Store in the horizontal position without stacking anything on top.
- Do not allow anyone to weigh without your supervision and instructions.
- When not using, turn the ON-OFF switch to OFF to conserve the battery.

8.1.3.2: Height Protocol

HEIGHT PROTOCOL
(last updated 4/20/21)

Prior to the appointment the participant is instructed to wear light clothing and informed that both height and weight measurements will be made with shoes off.

Materials

- Detecto PHR Portable Stadiometer
- Measurements will be made in centimeters

Procedure

Participant

- Shoes removed.
- Back to the height board, heels all the way to the back.
- Standing up straight with arms by sides (**Evaluate posture** to be sure participant is standing upright, flat-footed and looking straight ahead with head in the Frankfort horizontal plane, the horizontal plane which includes the lower margin of the bony orbit--the bony socket containing the eye--the most forward point in the supratragal notch--the notch just above the anterior cartilaginous projections of the external ear). See "Frankfort Plane for Measuring Body Height" diagram.

Research Staff

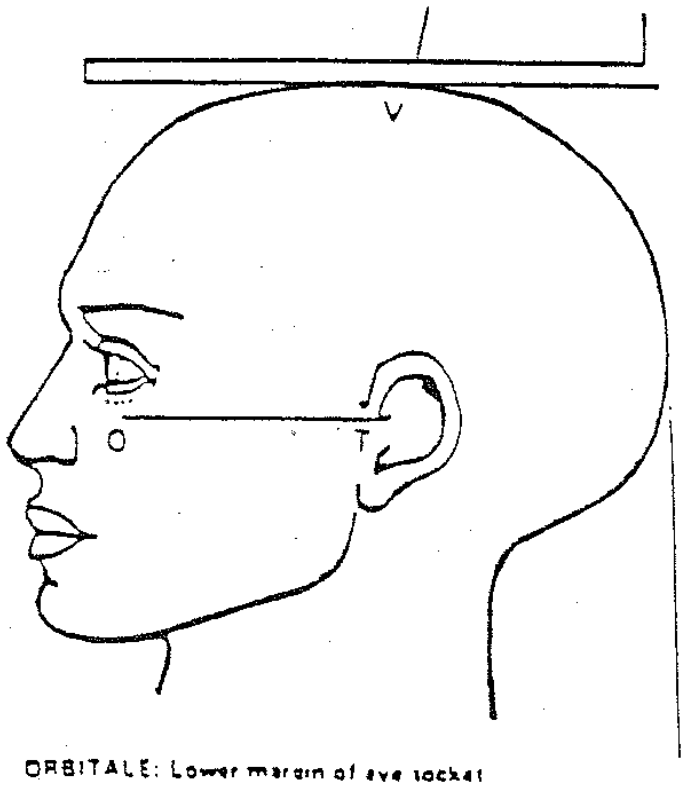
- Set up stadiometer following included instructions.
- Ensure headpiece is set-up with no offset. This model allows one to set up the stadiometer with an offset to accommodate the participant standing on a scale. However, participants will not be standing on a scale for this study, so the stadiometer should be set up without an offset.
- Lower the sliding headpiece onto the participant's head.

- Read the measuring scale of the stadiometer to the **nearest 0.1 cm** (Note: record all measurements in centimeters).
- Two measures are taken and recorded on measurement form. The participant steps off the stadiometer between measures. If the two measures **differ by >0.5 cm, a third measure is taken** and recorded.
- Height will be determined from the average of all recorded observations.

Notes:

- If the participant is breathing heavily enough to cause oscillations in the sliding headpiece, you must wait until s/he has settled down or ask to exhale and hold breath.
- If the participant's hairstyle interferes with measurement, be flexible and creative and get an accurate measurement (i.e. remove ponytail holders, measure hair height with tape measure, flatten hairdo, etc.).
- If the participant's buttocks cause an irregular or very unnatural posture when the heels are against the back of the board, allow standing with feet under hips (heels will not be against the back of the board in this case).

Frankfort Plane for Measuring Body Height



ORBITALE: Lower margin of eye socket

TRAGION: Notch above tragus of ear or at upper margin of zygomatic bone at that point

FRANKFORT PLANE: Orbitale tragion line horizontal

8.1.3.2: Blood pressure protocol

BLOOD PRESSURE PROTOCOL
(updated 4/20/21)

Warnings

- Do not share an electric outlet with another unit or electric appliance.
- Keep unit at least 5 feet from cell phones and computers.
- Unplug AC adaptor from electric outlet if unit is unused for an extended period of time.
- Do not inflate the cuff without being wrapped over the arm.
- Do not store unit under high temperature and high humidity.
- Transport the unit inside a car, not in the trunk.

Materials

- Omron Model HEM-907XL blood pressure monitor with small, medium, large, and extra large cuffs
- Tape measure

Note: The Omron cannot be used on arms with a circumference greater than 17" (large Omron cuff does not fit) or if the Omron is unable to take the measurement.

Procedure using the Omron

Set up

- Connect the AC adapter to the DC jack of the main unit and the electric outlet.
- Connect the air tube to the main unit by securing the air plug to the base of the air connector. This air tube is for use with the small, medium, and large cuffs only. The extra large cuff has its own non-detachable air tube for connection to the main unit.

Participant

- Should avoid caffeine, exercise, and smoking for at least 30 min before measurement.
- Invite participant to empty bladder before measurement.
- Shirt, blouse, or dress sleeves are loose fitting enough not to cause constriction of the arm even when pushed up for wrapping the cuff onto the bare arm. If too tight, the arm needs to be placed outside the sleeve.
- Remains quietly seated in a chair with both feet flat on the floor and back supported while measurements are taken. Ideally there would be a table or desk nearby to support the arm. Neither the participant nor the staff should talk during the 5 min rest period or during the measurement.

Research Staff

- Apply the cuff to the right arm of the seated participant. **All BP measurements are made using the <u>right arm</u> unless prevented by injury or deformity.** If the left arm is used it is indicated on the BP and Heart Rate (BPHR) form and subsequent BP measures are taken using the left arm. See "How to Apply the Cuff" diagram included with instrument.
  - Measure the circumference of the largest part of the upper arm and select the cuff according to arm size.
    - 7"-9" small
    - 9"-13" medium
    - 13"-17" large
    - 17"- 20"extra large
  - Connect the air tube and the cuff/bladder set by rotating the Luer connection.
  - Place the participant's hand with the palm upward.
  - Align the Artery Position Mark on the cuff with the brachial artery.
  - Wrap the cuff snugly using both hands and securely fasten it with the Velcro™ tape. The **lower edge of the cuff should be placed ½" to 1" above the inner side of the elbow joint**.
    - If the INDEX is positioned outside the RANGE, select the cuff suitable for the patient's arm circumference and wrap it again. Wrap the cuff so that you can insert only one finger between the cuff and arm. If the upper arm is more pyramid shaped rather than cylinder shaped it is necessary to wrap the cuff in a slightly diagonal fashion so that both the upper and lower borders of the cuff fit snugly against the skin.
  - Keep the **level of the cuff at the same level as the heart** during the measurement. The participant should sit upright in the chair, not slouching, with arm supported (e.g., resting on a desk or table).
- Take the blood pressure.
  - Push the ON/OFF (power) BUTTON to turn on the power.
  - Set the MODE Selector to "AVG". The monitor has been programmed to:
    - Wait 5 min. until the 1st measurement.
    - Take 3 measurements.
    - Wait 1 min. between measurements.
  - Set the P-SET (inflation level setting) Knob to "AUTO".
  - Push the HIDE button for non-display of results.
  - Push the START Button to begin measurement. The 5 minute countdown will be displayed. After 5 minutes the monitor takes the first measurement and then displays the results. One minute later the 2nd measurement is taken. The monitor takes the 3rd and final measurement after another minute.
    - To stop a measurement, push the STOP Button. The unit will rapidly deflate.
    - Do not push the START Button without first wrapping the cuff around the participant's arm.
    - If an error occurs during measurement, the monitor will automatically start measurement again. If a second error occurs, measurement will automatically stop.

        - If the monitor determines that the pressure value is not correct, an error display appears (Er1 to 9). In this case, refer to Page 28 of the manual and start the measurement again.
        - **If the blood pressure is unattainable using the monitor the measure will be missing. <u>Note this on the form</u>** by checking appropriate option 1) arm circumference greater than 17" 2) Omron unable to measure.
    - Push the HIDE button for display of results.

- Enter measurement results on the form.
    - After all 3 measurements are completed; the average value will be displayed.
    - Enter the average pulse rate (BPM) on the form.
    - Press the DEFLATION (deflation control) /Measurement Result Display Switch Button to display the 1st measurement for entering.
    - Press the same button again to display the 2nd measurement for entering.
    - Press the button again to display the 3rd measurement for entering.
- Push the ON/OFF (power) Button to turn off the power.
- Mark right/left arm and record your staff ID number on the form.

Blood Pressure Feedback Card

- Fill in the date and the average of the 3 readings.
- Check a blank for recommendation if appropriate.
- Give to participant and point out any checked recommendation.
- If blood pressure is > 140/90 we will recommend the following:
    - 140-159/90-99 [Check with your clinician within 2 months.]
    - 160-179/100-109 [Check with your clinician within 1 month.]
    - 180-219/110-129 [Check with your clinician within 1 week.] Also, staff will send results to primary care clinician.
    - Greater than 220/130 [Check with your clinician within 24 hours.] Also, staff will send results to primary care clinician.
    - *If the systolic and diastolic pressures are in different categories, follow the recommendations for the category with the shortest follow-up interval

8.1.3.3: Blood specimen collection protocol

Collection of Blood Specimens
(last updated 4/20/21)

This document outlines the lab protocols for this study. The lab work for this study will done by a commercial laboratory and is generally considered routine clinical lab tests.

- This lab work will be performed by LabCorp.
- Lab measurement time points: baseline, and 6 and 12 months follow-up.

- Lab tests to be done as outlined in the proposal
    - Hemoglobin A1c

Test Collection and Sample Stability Information (as outlined on LabCorp website)

Test	Code	Tube	Preparation	Vol.	Temp	Stable
Hemoglobin A1c	001453	Lavender top	Not needed	3 ml stated; ped tube may be used	Room	14 days

Collection Protocol

- Owing to participant burden of fasting labs, non-fasting labs will be collected.
- Complete LabCorp requisition and necessary tube labels as directed by LabCorp Note: lab for routine clinical lab tests will include identifying data including participants name and date. The specimens will be discarded according to the standard operating procedures of LabCorp.
- **Using standard phlebotomy technique, collect:**
    - 3cc lavender-top tube (Hemoglobin A1c)
        - No centrifugation needed.
    - Deliver to LabCorp (samples do <u>not</u> need to be refrigerated)

8.1.4: Questionnaire data

General protocol for collecting questionnaire data

- See above for outline of how eligibility, baseline, and follow-up questionnaires will be administered.

Questionnaire data will include data elements as describe in the measures table above. We will also collect the additional information described below.

Medication use

- Our objective is to assess medication use and change in medication use for the following conditions.
    - Diabetes
    - High blood lipids
    - High blood pressure
    - Obesity (FDA approved weight loss medication)
- To reduce participant burden, we will collect medication names, but not dosage or frequency of use
- We will distinguish between routine and PRN use

- If it is apparent that a participant is unsure about medications and assistance from a clinician is needed to better determine medication use, Dr. Berkowitz will be contacted to call the patient within 2 weeks of the visit to clarify medication use.

Adverse outcomes

- We do not anticipate adverse outcomes to be a consequence of the intervention.
- We will however assess for emergency room visits and hospitalizations at each follow-up questionnaire.
- We will categorize emergency room and hospitalizations as follows:
    - Acute myocardial infarction
    - Angina pectoris
    - Stroke
        - Thrombotic
        - Hemorrhagic
    - TIA
    - Heart failure
    - Coronary bypass surgery
    - Cancer
        - Type of cancer
    - Broken bone
        - Type of fracture
    - Light headedness or dizziness
    - Pneumonia
    - Other infection
        - UTI
        - Other
    - Hypoglycemia
    - Other
- We will also ask about incident cardiovascular events that did not result in a hospitalization or emergency department visit. These are:
    - Acute myocardial infarction
    - Angina pectoris
    - Stroke
        - Thrombotic
        - Hemorrhagic
    - TIA
    - Heart failure

8.1.5: Study measures returned to participants

Routine study measures that participants can interpret, will be returned to participants, including:

- Weight

- Height
- Body Mass Index
- Blood pressure
- Hemoglobin A1c

8.1.6: Study measures will also be sent to the participant's primary care clinician (PCP)

- The PCP will be notified about high blood pressure as outlined above in Section 8.1.3.2: above.

8.1.7: Data supplementation

To help supplement data in the case of data not being collected from study visits, hemoglobin a1c, systolic and diastolic blood pressure, and height and weight measurements will be extracted from electronic health records where available, along with the date of observation. These may be used if study visit data are missing and an equivalent observation was made within +/- 45 days of the target measurement date.

8.2. SAFETY ASSESSMENTS

As noted, this study is considered minimal risk. That said, the following safety procedures are included as part of this protocol.

- Because the intervention simply provides healthy food to those with T2DM and food insecurity, we anticipate few adverse events from the intervention itself. However, the study sample is likely to face some risks due to their overall circumstances. Key risks we anticipate are low blood sugar and the possible need for medication adjustment. Though we saw reductions in hypoglycemia during the pilot trial[67], those with food insecurity and T2DM are at high risk for this adverse event. During the baseline visit, study staff will discuss signs and symptoms of hypoglycemia, and remind participants to develop a treatment plan for this with their treating physician. Medical management of hypoglycemia will be under the purview of the participant's care provider. Referral to CS requires a care provider certification so all study participants will have a care provider. Over the course of the study, study staff will record hypoglycemic events and communicate them to the treating physician, to adjust treatment plans as needed. These calls will occur monthly, as part of the 'check-in' call 1-2 weeks after the lifestyle intervention session
- As part of the lifestyle intervention, an information sheet on safety while being active will be given to participants in the intervention group.
- If a participant loses more than 40 pounds at the 6 month follow-up, their study record will be reviewed by Dr. Berkowitz, who may contact the participant's PCP to determine if additional evaluation is warranted for underlying medical disease that may be contributing to weight loss.

8.3 ADVERSE EVENTS AND SERIOUS ADVERSE EVENTS

8.3.1 DEFINITION OF ADVERSE EVENTS

This protocol uses the definition of adverse event from 21 CFR 312.32 (a): any untoward medical occurrence associated with the use of an intervention in humans, ***whether or not considered intervention-related***.

8.3.2 DEFINITION OF SERIOUS ADVERSE EVENTS

For this study a serious adverse event includes, as outlined at: *https://www.hhs.gov/ohrp/regulations-and-policy/guidance/reviewing-unanticipated-problems/index.html*:

1. results in death;
2. is life-threatening (places the subject at immediate risk of death from the event as it occurred);
3. results in inpatient hospitalization or prolongation of existing hospitalization;
4. results in a persistent or significant disability/incapacity;
5. results in a congenital anomaly/birth defect; or
6. based upon appropriate medical judgment, may jeopardize the subject's health and may require medical or surgical intervention to prevent one of the other outcomes listed in this definition (examples of such events include allergic bronchospasm requiring intensive treatment in the emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse)

8.3.3 CLASSIFICATION OF AN ADVERSE EVENT

8.3.3.1 SEVERITY OF EVENT

For adverse events (AEs) not included in the protocol defined grading system, the following guidelines will be used to describe severity.

- **Mild** – Events require minimal or no treatment and do not interfere with the participant's daily activities.
- **Moderate** – Events result in a low level of inconvenience or concern with the therapeutic measures. Moderate events may cause some interference with functioning.
- **Severe** – Events interrupt a participant's usual daily activity and may require systemic drug therapy or other treatment. Severe events are usually potentially life-threatening or incapacitating. Of note, the term "severe" does not necessarily equate to "serious".

8.3.3.2 RELATIONSHIP TO STUDY INTERVENTION/EXPERIMENTAL MANIPULATION

All adverse events (AEs) will have their relationship to study procedures, including the intervention, assessed by an appropriately-trained clinician based on temporal relationship and his/her clinical judgment. The degree of certainty about causality will be graded using the categories below.

- **Definitely Related** – There is clear evidence to suggest a causal relationship, and other possible contributing factors can be ruled out. The clinical event, including an abnormal laboratory test result, occurs in a plausible time relationship to study procedures administration and cannot be explained by concurrent disease or other drugs or chemicals. The response to withdrawal of the study procedures should be clinically plausible. The event must be pharmacologically or phenomenologically definitive.
- **Probably Related** – There is evidence to suggest a causal relationship, and the influence of other factors is unlikely. The clinical event, including an abnormal laboratory test result, occurs within a reasonable time after administration of the study procedures, is unlikely to be attributed to concurrent disease or other drugs or chemicals, and follows a clinically reasonable response on withdrawal.
- **Potentially Related** – There is some evidence to suggest a causal relationship (e.g., the event occurred within a reasonable time after administration of study procedures). However, other factors may have contributed to the event (e.g., the participant's clinical condition, other concomitant events). Although an AE may rate only as "possibly related" soon after discovery, it can be flagged as requiring more information and later be upgraded to "probably related" or "definitely related", as appropriate.
- **Unlikely to be related** – A clinical event, including an abnormal laboratory test result, whose temporal relationship to study procedures administration makes a causal relationship improbable (e.g., the event did not occur within a reasonable time after administration of the study procedures) and in which other drugs or chemicals or underlying disease provides plausible explanations (e.g., the participant's clinical condition, other concomitant treatments).
- **Not Related** – The AE is completely independent of study procedures administration, and/or evidence exists that the event is definitely related to another etiology. There must be an alternative, definitive etiology documented by the clinician.

8.3.3.3 EXPECTEDNESS

Dr. Berkowitz will be responsible for determining whether an adverse event (AE) is expected or unexpected. An AE will be considered unexpected if the nature, severity, or frequency of the event is not consistent with the risk information previously described for the study procedures.

8.3.4 TIME PERIOD AND FREQUENCY FOR EVENT ASSESSMENT AND FOLLOW-UP

Solicited potential adverse events will occur at each follow-up measurement visit (6, and 12 months). They include the following and we will ask if evaluation for these events included emergency room visit or overnight hospital stay.

- Acute myocardial infarction
- Angina pectoris
- Stroke
    - Thrombotic
    - Hemorrhagic
- TIA
- Heart failure
- Coronary bypass surgery
- Cancer
    - Type of cancer
- Broken bone
    - Type of fracture
- Light headedness or dizziness
- Pneumonia
- Other infection
    - UTI
    - Other
- Hypoglycemia
- Other

Unsolicited adverse events (AE) or serious adverse events (SAE) may come to the attention of study staff during counseling visits, phone calls, or via other channels of communications. They will be categorized as above and entered into the appropriate follow-up questionnaire at the time of occurrence.

Information to be collected about AEs and SAEs includes event description, time of onset, study clinician's assessment of severity, relationship to study procedures (assessed only by those with the training and authority to make a diagnosis), and time of resolution/stabilization of the event. All AEs occurring while on study will be documented appropriately regardless of relationship. All AEs will be followed to adequate resolution.

Any medical or psychiatric condition that is present at the time that the participant is screened will be considered as baseline and not reported as an AE. However, if the study participant's condition deteriorates at any time during the study, it will be recorded as an AE.

Changes in the severity of an AE will be documented to allow an assessment of the duration of the event at each level of severity to be performed. Documentation of onset and duration of each episode will be maintained for AEs characterized as intermittent.

Study staff will record events with start dates occurring any time after informed consent is obtained until the last day of study participation.

8.3.5 ADVERSE EVENT REPORTING

Adverse events will be reported as follows:

SAEs that include the following and are 1) definitely, 2) probably, or 3) potentially related to the study will be reported to the Independent Safety Monitor and IRB within 1 working day:

- Results in death
- Is life-threatening (places the subject at immediate risk of death from the event as it occurred)

Other SAEs and AEs will be reported to the Chair of the DSMC (by the 15th of each month for the preceding month) and will be summarized for review at each meeting of the DSMC.

Reports of adverse events to the UNC IRB will follow their policy as outlined below:

1. Do not wait to submit a reportable event to the IRB until you have all the information about the event. Report the event promptly and follow-up with additional information as it becomes available.
2. A UPIRSO (Unanticipated Problem Involving Risks to Subjects or Others) that is also a Serious Adverse Event must be reported ASAP, but no later than one (1) week from the time you become aware of the event.
3. All other UPIRSOs must be reported ASAP, but no later than two (2) weeks from the time you become aware of the event.

8.3.6 SERIOUS ADVERSE EVENT REPORTING

As outlined above in Section 8.3.5

8.3.7 REPORTING EVENTS TO PARTICIPANTS

Participants will be aware of events we consider AEs and SAEs as they will be reported to us by participants. Given the low risk nature of this study, we do not anticipate reporting summary information on adverse events to participants. We do not consider elevations in hemoglobin A1c to be incidental as they likely reflect an underlying biologic process that PCPs are familiar with and able to respond to in an appropriate fashion.

8.3.8 EVENTS OF SPECIAL INTEREST

Not applicable

8.3.9 REPORTING OF PREGNANCY

Pregnancy: Study participants who become pregnant will be removed from the study as gestational diabetes is a separate condition from type 2 diabetes, the focus of the study. However, we do not anticipate any harms from continuing the study if the patient is unaware she is pregnant.

8.4 UNANTICIPATED PROBLEMS

8.4.1 DEFINITION OF UNANTICIPATED PROBLEMS

This protocol uses the definition of Unanticipated Problems as defined by the Office for Human Research Protections (OHRP). OHRP considers unanticipated problems involving risks to participants or others to include, in general, any incident, experience, or outcome that meets **all** of the following criteria:

- Unexpected in terms of nature, severity, or frequency given (a) the research procedures that are described in the protocol-related documents, such as the Institutional Review Board (IRB)-approved research protocol and informed consent document; and (b) the characteristics of the participant population being studied;
- Related or possibly related to participation in the research ("possibly related" means there is a reasonable possibility that the incident, experience, or outcome may have been caused by the procedures involved in the research); and
- Suggests that the research places participants or others at a greater risk of harm (including physical, psychological, economic, or social harm) than was previously known or recognized.

8.4.2 UNANTICIPATED PROBLEMS REPORTING

The investigator will report unanticipated problems (UPs) to the reviewing Institutional Review Board (IRB) and to the Independent Safety Monitor. The UP report will include the following information:

- Protocol identifying information: protocol title and number, PI's name, and the IRB project number
- A detailed description of the event, incident, experience, or outcome
- An explanation of the basis for determining that the event, incident, experience, or outcome represents an UP

- A description of any changes to the protocol or other corrective actions that have been taken or are proposed in response to the UP

To satisfy the requirement for prompt reporting, UPs will be reported using the following timeline:

- UPs that are serious adverse events (SAEs) will be reported to the IRB and to the DSMC ASAP but no later than one week of when the investigator becoming aware of the event.
- Any other UP will be reported to the IRB and to DSMC ASAP but not later than within 2 weeks of the investigator becoming aware of the problem

8.4.3 REPORTING UNANTICIPATED PROBLEMS TO PARTICIPANTS

N/A as participants will be aware of unanticipated problems

9 STATISTICAL CONSIDERATIONS

The statistical analysis plans will be consistent with guidance of "Statistical Principles for Clinical Trials: Addendum on Estimands and Sensitivity Analysis in Clinical Trials" (ICH E9-R1), the CONSORT Statement[121], ICMJE recommendations[122], and the 2016 and 2019 statements of the American Statistical Association[123,124] with associated commentaries in Nature.[125,126]

9.1 STATISTICAL HYPOTHESES

This single-site, open-label, Phase II, community-based randomized controlled explanatory trial to test the efficacy of a medically tailored meal (MTM) interventions for adults with food insecurity and T2DM, compared with a group that receives usual care + a monthly food subsidy. Our study aims are:

Aim 1 (QUAN): To determine the efficacy of a medically tailored meal intervention program for individuals with type 2 diabetes in a 6-month randomized clinical trial (n=200)

H1a. Primary outcome. Hemoglobin A1c: We hypothesize that the MTM group will have 0.5% lower hemoglobin A1c at the end of treatment (6 months, primary analysis) and 6 months after treatment ends (12 months, secondary analysis), compared with the 'usual care + food subsidy' group.

H1b. Patient-reported outcomes: We hypothesize that the MTM group will have less food insecurity and hypoglycemia, and greater improvements from baseline in diabetes distress, diabetes and diet self-efficacy, and health-related quality of life at the end of treatment (6 months) and 6 months after treatment ends (12 months), compared with the 'usual care + food subsidy' group.

Aim 2 (qual and quan): To assess why, how, and under what circumstances MTM participants A) achieve improvements in T2DM outcomes and B) maintain these improvements after the intervention.

The statistical analyses for these aims are:
Aim 1. We will evaluate the efficacy of the MTM intervention, relative to the control regimen, for the target population in terms of measures of the following outcome variables:
(1a) Hemoglobin A1c at the end of the 6-month treatment period (primary outcome) and measured six months after the end of the treatment period (secondary outcome).
(1b) Food insecurity, hypoglycemia, diabetes distress, diabetes self-efficacy, diet self-efficacy, and health-related quality of life measured at the end of the treatment period and six months later.
For each of the Aim-1 measures, the population parameters (estimands) of primary interest are mean levels (indexed by regimen and occasion) and differences among those mean levels.

Aim 2. We will assess why, how, and under what circumstances MTM-treated participants achieve and maintain improvements in the outcome measures.
(2a) Qualitative interviews at the end of MTM treatment and 6 months later will be used to investigate cognitive ('why?'), procedural ('how?'), and contextual ('under what circumstances?') dimensions of the response to the MTM treatment regimen. These will help identify potential mechanisms and moderators of treatment effects, such as race/ethnicity, and inform future modifications.
(2b) Quantitative analyses will be used to investigate the 'how' by examining putative mediating effect of changes in diet and weight, compensatory strategies, and the psychological factors. Moderating effects of patient characteristics will also be examined.

Qualitative data will be analyzed qualitatively, not statistically, as described below.

9.2 SAMPLE SIZE RATIONALE

The target sample size is N = 200 enrolled participants. We anticipate a drop-out rate < 10% and thus anticipate that n = 190 participants will complete the study. We observed a low rate of attrition in our pilot study, but we recognize that the longer time frame of this trial means attrition is more likely. The primary considerations in choosing our proposed sample size included the anticipated frequency of missing data, anticipated levels of precision of key estimators, anticipated levels of power of a hypothesis test regarding HbA1c, and anticipated availability of participants.

HbA1c Treatment Effect. In terms of change in HbA1c at 6 months, we hypothesize that the treatment effect is 0.5% or greater in the target population. This magnitude of effect is consistent with our pilot results[67] and is clinically meaningful: the American Diabetes Association and the European Association for the Study of Diabetes use this threshold to categorize glucose-lowering efficacy as low versus intermediate,[127,128] and the Food and Drug Administration accepts an even lower threshold of 0.3%.[129]

Anticipated Precision: HbA1c. We assume the population standard deviation (SD) of HbA1c (%) is near 1.1% as observed in previous samples of MTM recipients.[78,79] It follows that the standard error (SE) for a sample mean observed in n=95 participants is expected to be SE = 0.1% with corresponding 95%CI [estimate ± 0.2%]. The treatment effect estimated by the difference between the two sample means would have SE = 0.15 and a 95%CI of the form [estimate ± 0.29%].

Anticipated Power: HbA1c. For a test of H_o "the 6-month treatment effect is zero in the target population", we have calculated power levels. The table lists power estimates for varying levels of attrition. This table is assumes a population treatment effect of 0.5% or greater, a SD of 1.1%, and 1:1 randomization. With a starting sample size of 100 per arm (n=200 total), we anticipate adequate power across a range of plausible levels of attrition.

Table: Power for Detecting ≥0.5% Difference in HbA1c

Starting Sample	% Attrition	Ending Sample	Power
200	0	200	89%
200	5	190	88%
200	10	180	86%
200	20	160	82%
200	23*	128	80%
200	30	120	76%

* largest amount of attrition that maintains 80% power

9.3 POPULATIONS FOR ANALYSES

Two analysis population will be defined for this study:

- *Intent-to-Treat (ITT) Population (primary population)*: In the primary analyses we will include all participants who were randomized, analyzed according to the treatment regimen to which they were assigned by randomization, regardless of extent of intervention actually received. To cope with missing data values in analyses of the ITT cohort's data, we anticipate that we will use methods such as linear mixed-effects models or multiple imputation (MI) methods for the primary analyses. See further details below.
- *Per Protocol (PP) Population (secondary population)*: This population will include data from all participants up until they stop following the protocol for the intervention they were assigned to, at which point their data will be censored. For the MTM group, censoring will occur if > 4 weeks of meal deliveries are missed, or if > 1 lifestyle session is not completed. For the Food Subsidy group, censoring will occur if no portion of the food subsidy is spent for > 2 consecutive months. For both groups, data will be censored for analyses in which outcome data are not available.
- *Intervention Cohort*: As described in the appropriate section, some qualitative analyses for Aim 2 will rely on information obtained from the intervention cohort only.

9.4 STATISTICAL ANALYSES

9.4.1 GENERAL APPROACH

Statistical analysis plans: In the quantitative analyses for Aim 1 and Aim 2, the main results will be obtained using appropriate model-based methods: e.g., for each regimen and for differences between the regimens, point estimates of central tendency and variance will be obtained along with 95% confidence intervals to indicate precision. To guide our level of trust in the main results, sensitivity analyses will be performed to assess the robustness / sensitivity of the main results to reasonable perturbations of the choices of assumptions and methods used. Examples of the sensitivity analyses include adjustment for additional covariates or interactions, analysis of residuals and influential observations, exclusion or inclusion of questionable / extreme data values. Our plans also include roles for hypothesis generation in exploratory analyses[130] and for descriptive graphical and tabular methods to characterize participants, visualize data, and examine relationships among variables.

In the qualitative analyses for Aim 2, we will examine the cognitive, procedural, and contextual dimensions of treatment response among participants assigned to the intervention regimen. This investigation will help us understand potential mechanisms, mediators, and moderators of treatment effects, and will inform future modifications of the intervention.

9.4.2 ANALYSIS OF THE PRIMARY ENDPOINT(S)

The primary analyses for Aim 1 will be conducted using the ITT cohort and will be repeated secondarily using the PP cohort (using appropriate methods that account for non-random exit from the study protocol). We will characterize and compare the two treatment regimens in terms of effect on HbA1c (primary outcome) and the additional outcomes as described in the measures table.

For HbA1c (%) measured at baseline (t=0), t=6 months, and t=12 months, the analysis will rely on a constrained longitudinal data analysis (cLDA) model[131,132].

The model will include terms for treatment regimen (MTM vs. food subsidy), time (baseline, 6 month, or 12 month visit), and their interaction. Using the cLDA approach, and given random treatment assignment, the baseline mean will be assumed to be equal between groups. To improve precision, the models will include baseline covariates. To permit valid estimation of marginalized means from models with covariates, treatment regimen will be interacted with baseline covariates as well. The comparison of primary interest is the mean level of the outcome at 6 months under the MTM exposure – the mean level of the outcome at 6 months under the food subsidy exposure. To provide marginal results from models containing covariates, we will express results as least-squares means and their contrasts.[133]

The cLDA model provides several advantages: it allows inclusion of participants with incomplete data, it is more efficient, it avoids conditioning on the baseline score, and it facilitates modeling the components of variance.[131,132] The model will be fitted under the assumption that the variance-covariance matrix is unstructured (3x3). Reasonable variations on the model assumptions will play an important role in the set of sensitivity analyses.

Relatedly, formal assessments of mediation will be addressed as part of the Aim 2 analyses.

Coping with Missing Data: Best practices for dealing with incomplete data will depend on the causes of the missing values and the amount of missing information. We will document the reasons for missing values and attempt to obtain follow-up data for all enrollees who discontinue treatment.

As an additional way to address missing data, we will include measures obtained as part of routine clinical care (e.g., an Hba1c result obtained in a participant's primary care clinic within the study period). If a participant is missing a study collected outcome, it may be supplemented with a measurement obtained from the electronic health record if the measurement was made within 45 days (before or after) the target data collection point (based on the participant's initial data collection visit).

Depending on the pattern of missingness encountered, we expect to use multiple imputation (MI) methods, particularly multiple imputation with chained equations, to cope with missing data values. MI yields multiple complete copies of the analysis dataset; however, the amount of information in the data that is available to the analysis is unchanged by the use of MI. MI does not recreate or restore information that was lost due to missing data. However, under a missing-at-random assumption, MI can help correct for biases if measured data, such as the treatment assignment group or baseline covariates, is associated with the missing outcomes. This may be particularly useful if there is differential loss to follow-up between groups.

Logical consideration based on the reasons for missing values will be used to assess whether the missing-data mechanisms are ignorable (missing completely at random, MCAR; or missing at random, MAR) or not-ignorable (missing not at random; MNAR).[137,138] Sensitivity analyses will include comparisons of the respondents and non-respondents in terms of baseline characteristics and the outcome measures.

Interpretation of Estimates and Inferences: The purpose of Aim 1 is to provide stakeholders with information regarding the magnitudes of MTM treatment effects relative to those of the food subsidy treatment regimen. The most important results will be point- and interval- estimates of treatment effects for each outcome variable. We are not seeking to test an overall composite (null) hypothesis spanning the several outcome variables. For example, the estimates and inferences for HbA1c and the PRO measures in Aim 1 will be interpreted as separate; a family-wise type 1 error rate is not of interest. In contrast, multiple-comparisons procedures will be used in when testing composite null hypotheses such as "the five subpopulations do not differ in terms of the treatment effects" in sub-group analyses; for example, we may use the false-discovery rate approach in these cases.[139,140]

Use of P-values. By design, statistical test procedures are incapable of establishing that the null hypothesis is true. P values can only provide evidence against the hypothesis being tested. Tests results (p values) that are deemed to be not statistically significant are entirely inconclusive and no conclusion

about the null hypothesis can be drawn from the test; this is true even if the corresponding confidence interval is narrow and thus highly informative; this is true regardless of the sample size. Consistent with the guidance of the American Statistical Association, P values will be reported and will not be dichotomized.[123,124]

9.4.3 ANALYSIS OF THE SECONDARY ENDPOINT(S)

For the other outcome measures in Aim 1, (e.g., food insecurity, hypoglycemia, health-related quality of life, diabetes distress, diabetes self-efficacy, diet self-efficacy, medication adherence, cost-related medication underuse, food-medication trade-offs, and depressive symptoms), longitudinal analysis methods very similar to those used for HbA1c will also be used. Regression models will use response distributions and link functions appropriate to the type of variable being analyzed (e.g. binomial response and logit link for binary outcomes, Poisson response and log link for count outcomes, etc).

Again, the ITT cohort will be used in the main analyses of the secondary outcome variables. Comparable analyses using the PP cohort will play an important role in the sensitivity analyses. Per Protocol analyses may use methods that account for non-random exit from the study protocol, such as 'G-methods'.[136]

If the above methods prove infeasible owing to data-structure issues, alternative methods may need to be selected. If this occurs, reasons for any changes will be documented.

Mediation and Effect Modification Analyses
We will explore the roles of potential mediators and moderators of the treatment effects.

Effect Mediators (variables in the causal pathway). Given existing empiric evidence, our theoretical model depicts 3 key pathways where MTM may exert an effect to improve diabetes outcomes: diet, compensatory behaviors, and psychological factors. The above measures table lists quantitative data we will collect to enable investigation into whether these factors serve a mechanistic, or mediating role, as well as whether these factors serve as effect modifiers. Because treatment assignment was random, we will use a causal mediation framework for these analyses.[141–146]

Effect Modifiers (baseline variables that influence treatment effects). To examine possible effect-modification by relevant contextual factors, we will fit models similar to those used in Aim 1 analyses, but that include baseline covariates that are putative effect modifiers along with terms representing their interactions with treatment regimen. Putative effect modifiers will include biologic sex, race/ethnicity, participation in other food insecurity interventions such as SNAP, and household size. In these analyses we will attempt to account for potential confounding introduced by observational sub-population comparisons.[147] The results obtained will be used to generate new hypotheses.[130]

Qualitative Analyses

Because of the embedded mixed methods design, qualitative analyses are an important part of the overall analysis plan for the proposal as they will be used to better under the quantitative analyses. Below we detail the specific areas focused on in the qualitative investigations and the analysis of qualitative data.

Interview content: Qualitative data will come from interviews conducted at approximately 6 months (the end of the intervention, Part A) and at 12 months (6 months after the end of the intervention, Part B). Sampling for the participants who will take part in these interviews is described in the Research Strategy. Separate interview guides will be created for part A and part B. Both will be structured around exploring the cognitive ('why?'), procedural ('how?'), and contextual ('under what circumstances?') dimensions of participants' response (if any) to their study arm. A particular area of interest will be the experiences of differing racial/ethnic groups, to inform efforts to improve racial/ethnic disparities in T2DM outcomes. The interview guide will emphasize open-ended and non-leading questions and statements.

· ***Cognitive:*** Questions in this domain will explore why MTM participation helps with T2DM self-management (or does not). At 6 months, interviewers will probe the perceived impact of MTM participation on factors important for maintaining an optimal HbA1c and T2DM self-management. In the nutritional pathway this will include the impact of the lifestyle intervention component of MTM on HbA1c and self-management. In the compensatory pathway this will include how MTM fits into other aspects of T2DM management participants may have used (e.g., glucose lowering medication), and how MTM relates to other food insecurity interventions participants may have used (e.g., SNAP). In the psychological pathway this includes the impact of stress, changes in confidence to manage diet and improve (or maintain) HbA1c and T2DM self-management. Further, we will explore the perceived importance of T2DM self-management, readiness and motivation to change, and time-orientation (e.g., the value of preventing complications far in the future). At 12 months, interviewers will probe: how participants coped with transitioning off of MTM and how T2DM management stress, confidence, and skills changed following MTM participation.

· ***Procedural:*** Questions in this domain will explore how MTM improved (or not) their HbA1c and T2DM management behaviors more broadly. A) At 6 months, interviewers will probe: participants' use of MTM and lessons learned in the educational sessions; any 'modeling' function of the meals for healthy eating, the extent to which MTM participation lessened trade-offs with other health-related social needs (such as by freeing up resources for improved medication adherence); and how the intervention promoted healthy behaviors, such as increasing physical activity. B) At 12 months, the focus will be on assessing why these behaviors increased, waned, or remained stable after MTM.

· ***Contextual:*** Questions in this domain will explore the individual and socioeconomic circumstances under which participants found MTM participation helpful (or not) for T2DM management. At both 6 and 12 months, interviewers will probe the ways in which individual material circumstances (e.g., housing status, food security, personal safety), individual comorbidities, and social milieu characteristics, such as household composition (e.g., young children in the home), care-taking responsibilities, and participation in other health-related social need assistance programs (e.g., SNAP), influenced A) on-treatment and B) post-treatment changes to study outcomes. We will also probe how MTM fits within

the broader social context of participants' lives in order to enhance the ability to disseminate and implement MTM, if it proves effective.

Analysis: All interviews and record lifestyle intervention sessions will be transcribed verbatim and analyzed using NVivo 11. Data analysis will be based on a grounded theory approach[148] and will occur concurrently with data collection to ascertain when thematic saturation is reached for each source type and sampling stratum. Two trained study staff members, working in parallel, will begin by open-coding the transcripts to form initial categories using an inductive approach.[149] During weekly team meetings, constant comparison[150,151] will be used to iteratively refine categories, discuss discrepancies until high inter-rater reliability (K≥0.80) is achieved, and to organize emergent major and minor themes. Analyses will compare findings across treatment response profiles to generate insights about the reasons for response variability. Consistent with the embedded-experiment mixed methods design[152], the qualitative data will be used to enhance the understanding of the RCT outcomes.[153] Qualitative findings will be used to provide a more nuanced picture of why and how the intervention produced the outcomes it did. Further, qualitative analyses will be used to examine heterogeneous treatment effects, to refine conceptual frameworks, to share lessons learned about designing and deploying an MTM intervention, to inform modifications to the intervention, and, if effective, to inform implementation plans.

9.4.4 SAFETY ANALYSES

During the study, data-captured adverse events will be coded by the principal investigators.
At the end of the study, each type of event (by seriousness, severity, relationship to study) will be descriptively summarized in frequency tables by treatment regimen (including only a single occurrence of any distinct type of event for any participant). Number of ER visits and hospitalizations will also be descriptively summarized by treatment regimen. No inferential statistics (p-values or confidence intervals) will be used for comparing safety data between treatment regimens.

9.4.5 BASELINE CHARACTERIZATION OF PARTICIPANTS

We will summarize baseline data (i.e., pre-randomization data) for the enrollees. Measures of central tendency and dispersion for continuous and certain discrete variables will include means, standard deviations, medians, minima, and maxima. Categorical data will be summarized with frequencies and percentages. Some continuous variables may also be grouped into categorical levels and evaluated in frequency tables. Inferential statistics (i.e., p-values and/or confidence intervals) will not be used when comparing the treatment arms on baseline variables.

9.4.6 PLANNED INTERIM ANALYSES

No interim analyses of outcome measures will be performed. (As a separate issue, adverse events will be continuously monitored and reported to ensure safety of participants.)

9.4.7 SUB-GROUP ANALYSES

We will treat all findings from sub-group analyses as exploratory. Further, we may not have statistical power for these comparisons. Finally, an explicit purpose of the qualitative work undertaken in the study is to determine potential mediators and effect modifiers that were not anticipated at the time the study was designed. Thus, pending the results from this work, we may conduct sub-group analyses not specified a priori. We will again regard results from any of these analyses as exploratory.

9.4.8 TABULATION OF INDIVIDUAL PARTICIPANT DATA

Individual participant data will not be published.

9.4.9 EXPLORATORY ANALYSES

Exploratory analyses will investigate relationships of the following longitudinal outcome variables to treatment regimen: questionnaire data as listed in the measures table and not indicated as secondary outcomes, blood pressure, weight loss, inpatient admission and emergency department use, These exploratory examinations of the data will yield new hypotheses.[130]

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATION

10.1 REGULATORY, ETHICAL, AND STUDY OVERSIGHT CONSIDERATIONS

10.1.1 INFORMED CONSENT PROCESS

10.1.1.1 CONSENT/ASSENT AND OTHER INFORMATIONAL DOCUMENTS PROVIDED TO PARTICIPANTS

Consent forms describing in detail the study intervention, study procedures, and risks will be given to the participant prior to the enrollment call. During the enrollment call, a trained study staff member will go over the consent form, reading it aloud and making no assumptions of literacy. The participant will be asked to provide verbal consent, which will be documented electronically using REDCap. Consent materials are included below:

University of North Carolina at Chapel Hill
Consent to Participate in a Research Study

Adult Participants

Consent Form Version Date: March 3, 2021
IRB Study # 20-2847
Title of Study: Food is Medicine: A Randomized Clinical Trial of Medically Tailored Meals For Individuals with Type 2 Diabetes Mellitus and Food Insecurity
Principal Investigator: Seth Berkowitz
Principal Investigator Department: Medicine-Internal Medicine
Principal Investigator Phone number: (919) 966-2276
Principal Investigator Email Address: seth_berkowitz@med.unc.edu
Funding Source and/or Sponsor: National Institutes of Health (NIH)

Study Contact Telephone Number: (919) 843-6637
Study Contact Email: kathier1@email.unc.edu

Commented [BSA1]: Update to final consent form once enrollment closed

_____
CONCISE SUMMARY

Food insecurity, or not being able to afford healthy food, makes it harder to manage diabetes. The best way to treat food insecurity is unknown. This study will compare two programs to address food insecurity for people with diabetes; home delivery of medically tailored meals (MTM), or a subsidy that can be used to purchase healthy foods. MTM includes counseling sessions about how to eat more healthy foods (called lifestyle counseling). Those who choose to be a part of the study will be randomly assigned, like flipping a coin, to one of the two programs. Both programs will last for 6 months. There will be three study sessions where we measure things important for the study by having you answer surveys and you have blood drawn. One sessions will be at the start of the study, a second at 6 months after the start of the study, and a 3rd one 12 months after the start of the study, 6 months after the program ends, to see how you are doing after the intervention is over.

This is a low-risk study. The intervention simply provides healthy food, or money to buy healthy food. There may be minor pain from the blood draws at each measurement visit. Taking part does require a time commitment for study sessions calls and counseling calls. We expect participants in both study groups will have reduced food insecurity and consider this a benefit of taking part in the study.
What are some general things you should know about research studies?
You are being asked to take part in a research study. Participating is voluntary.
You may choose not to participate. You may also leave the study at any time, for any reason. There will be no penalty.

Research studies are done to learn new information. This new information may help people in the future. You may not receive any direct benefit from being in the research study. There also may be risks to being in research studies. Deciding not to be in the study or leaving the study before it is done will not affect your relationships with the researcher, Community Servings, or the University of North Carolina-Chapel Hill.

Details about this study are below. It is important that you understand this information so that you can make an informed choice about whether or not you would like to be in the study.

You have a copy of this consent form. As we go through the consent please ask any questions you may have at any time. Your understanding of the study is important.

What is the purpose of this study?
Having enough healthy food is a common problem for people with diabetes. This makes it hard to follow the diet needed to manage diabetes. In this study, we are comparing two programs to increase access to healthy food. We want to see if one helps improve blood sugar better than the other. The first program will send meals that are made for people with diabetes to their homes. People in this group will also get counseling to learn how to improve blood sugars with food and exercise. The second program will send people a subsidy for healthy foods in the form of a grocery store gift card. We will ask that this card be used to buy healthy foods. We will look at bloodwork to see if there is a difference in the blood sugar levels of the two groups. We will also use surveys to see if the programs affect people's lives in other ways. An example might be by reducing stress. Finally, we will use interviews to find out how the programs may have affected diabetes management.

You are being asked to be in the study because you have diabetes and have said you have trouble getting enough healthy food.

Reasons you should not be in this study:
You should not be in this study if:
- You are or have been in a diabetes, nutrition, or weight study in the last 12 months.
- Someone in your home is a part of this study. Only one member of each household may participate.
- You are considering bariatric surgery in the next year.
- You have had bariatric surgery in the past.
- You are homeless.
- You do not have a place to store food.
- You do not have a telephone.
- You are pregnant, breastfeeding, or intend to get pregnant in the next year.
- You have a history of cancer.
- You have stage 4 or 5 kidney disease.
- You have misused drugs or alcohol in the past 2 years.
- You have a psychiatric illness that will make you unable to participate in the study's activities.
- You intermittently use medications (like steroids) that are likely to raise your blood sugar

How many people will take part in this study?
Approximately 200 people will be in this study.

How long will your part in this study last?
You will be in the study for about 1 year.

What will happen if you take part in the study?
First, study staff called you to see if you were interested in the study. If you were, we mailed or emailed you this consent form. Now, we are calling to review the consent form with you and to answer any questions you have. After we have finished reviewing this form, you will decide if you would like to join the study. If yes, you will give verbal consent over the phone.

If you decide to participate, we will continue this phone call to start your first study session. You will answer questions about things like what you eat, what types of activities you do and other questions about your health. This call should last about 1 hour.

After the phone call, a nurse will come to your home to check your weight, height, and blood pressure. The nurse will also collect a blood sample to check your hemoglobin A1c. This is a test of your blood glucose over the past 3 months. We will report the results of these test to you and your primary care clinician. This nurse visit will take about thirty minutes.

When the nurse comes to your home, he or she will be wearing personal protective equipment, sometimes called "PPE". This is to protect both of you from the possible spread of infection. Their PPE will include a head cap, face shield, goggles, mask, suit, gloves and shoe covers. We ask that you also wear a mask while the nurse is in your home.
After this first study session, a computer program will randomly assign you, like flipping a coin, to receive either the Medically Tailored Meal Program or the Food Subsidy Program. You will then receive a call from a member of the study team to tell you which program you are in and go over next steps.

Medically Tailored Meal (MTM) Program
If you are assigned to this program, you will receive a Community Servings' meal delivery to your home each week for six months. The delivery will have five dinners, five lunches, milk, cereal, yogurt, fresh fruit and snacks. The meals are medically tailored. This means they are made to help improve your blood sugars. You will also receive counseling on ways to improve your blood sugar through lifestyle change. This will focus on healthy eating and staying active. You will have a coaching session over the phone that lasts about 45 minutes every month, for six months. You will also have a brief (about 15 minute) phone check-in one to two weeks after each coaching session. This will be a total of twelve phone sessions in six months.
The counseling sessions will be recorded for quality improvement. This will let us check to see how our counselors are doing. We can then work with them to make sure they are doing their best. These recorded sessions will be kept as electronic files but without information that could be used to identify you (other than your voice).

Food Subsidy Program
If you are randomized to the food subsidy program, you will be sent one $40 grocery store gift card each month, for six months. These cards are to be used to purchase healthy foods. We will send you a booklet with information on foods to purchase. We will also call you once a month to find out what types of foods you have bought with the card and to see how you are doing.

Follow-up Study Sessions
You will have two follow-up study sessions. The first will be about 6 months after you start the study. The second will be about 12 months after you start the study. They will be the same as your first study session. You will be called and asked to answer questions over the phone. This call will take about 1 hour. Then you will have a nurse visit to your home. The nurse will check your weight, blood pressure, and take blood to check your hemoglobin A1c. Hemoglobin A1c is a measure of your blood sugar. When the nurse comes to your home, he or she will be wearing personal protective equipment, sometimes called "PPE". This is to protect both of you from the possible spread of infection. Their PPE will include a head cap, face shield, goggles, mask, suit, gloves and shoe covers. We ask that you also wear a mask while the nurse is in your home.
You may also be asked to be interviewed about your experiences with the study. These interviews are optional. They would be completed after the 6 and 12 month nurse visits.

What are the possible benefits from being in this study?
Research is done to gain new knowledge. By being in this study you may have more access to healthier foods and better blood sugar control. These are not guaranteed benefits.

What are the possible risks or discomforts involved from being in this study?
There are not many possible risks and discomforts from being in this study. Those that are possible are listed below.

- We do not think there is risk to you from the food, money to buy food, or dietary advice given as part of the program.
- You may choose to lose weight as part of this program. If you do not eat enough, you could lose weight too fast. We will ask you to check your weight often. This way we will know if you are losing too much weight too quickly.
- Doing more physical activity is not the main focus of this study. However, the medically tailored meal program will recommend you do moderate physical activity. Those who increase their level of physical activity may experience minor muscle pain. This is common, affecting more than 50 in a 100 people. Minor muscle pain will go away. This type of activity rarely, less than 1 in a 100, causes serious health problems such as chest pain.
- Changing how you eat could lead to lower blood sugar. Sometimes this could include hypoglycemia. This has occurred in less than 1 in a 100 people in our past studies. We will provide information on how to reduce the risk of low blood sugar.
- For those with high blood pressure, changing how you eat could lead to lower blood pressure. Blood pressure that is too low can cause dizziness and even falls. It is very rare that this could be a serious risk, (less than 1 in a 100 people). Typically, the blood pressure is lowered slowly. This will allow your doctor time to observe the lower blood pressure and reduce any blood pressure medicine you may be taking.
- A blood sample will be collected by trained staff three times during the study. The risk of minor pain is very common. It has been seen in more than half of people. Bruising does not happen often (about 1-10 of every 100 persons) and the risk of infection or fainting is rare (fewer than 1 in 100).

- In all studies, there is a very slight chance of loss of privacy. This means the risk that others may see your study information. As we describe in more detail below, we will do all we can to make sure this does not happen.

Also, there may be other risks we did not list here, including uncommon or previously unknown risks. Because gestational diabetes is a different medical condition than type 2 diabetes, if you become pregnant during the study you will be removed from the study. You should report to the research team any problems that may be due to this study.

What if we learn about new findings or information during the study?
You will be given any new information gained during the study that might affect your willingness to continue participating.

Will I receive any other clinical results?
You will receive all clinical results. We will send you the results of your blood work, for example, your hemoglobin A1c. We will also send you any other measurements such as blood pressure and weight we obtain.

How will information about you be protected?
Your study bloodwork, your hemoglobin A1c, will be sent to a commercial lab for analysis (LabCorp). The bloodwork will have your name on it when it is sent. These samples are typically thrown away 3 days after collection. The results of your bloodwork along with your study surveys and interviews will be stored with your study ID number and your first and last initial. They will NOT be recorded with your name. We call this de-identified data. No one other than study staff will be able to connect your name and study ID as we will follow standard procedures to protect the privacy of research data.

We may use your de-identified data, as described above, in future research without additional consent. However, in some cases, the Institutional Review Board (called IRB and described below) may require that you be re-contacted and asked for your consent/permission to use your data in a specific research study. You have the right, at that future time, not to participate in any research study for which your consent is sought.

When writing reports or publishing articles participants will not be identified by name. Although every effort will be made to keep research records private, there may be times when federal or state law requires the disclosure of such records, including personal information. This is very unlikely, but if disclosure is ever required, UNC-Chapel Hill will take steps allowable by law to protect the privacy of personal information. In some cases, your information in this research study could be reviewed by agents of the University or research sponsors for purposes such as quality control or safety.

By providing informed consent, you agree that some of the information generated by participating in this study and/or a copy of this consent form may be included in your medical record. You also agree

that this information may be viewed by other physicians or caregivers who provide healthcare services to you. This will allow the doctors caring for you to know what tests you may be receiving as a part of the study. They will then know how to take care of you if you have other health problems or needs during the study.

If you are put in the medically tailored meals program you will participate in phone coaching sessions. These sessions will be recorded for review. These recordings will be identified by your study ID number, not by your name.

Part of the study involves optional interviews that will be recorded for review. You do not have to participate in these interviews to participate in the rest of the study, and we will ask you before each interview whether you want to participate in it.

What is a Certificate of Confidentiality?

This research is covered by a Certificate of Confidentiality. With this Certificate, the researchers may not disclose or use information, documents or biospecimens that may identify you in any federal, state, or local civil, criminal, administrative, legislative, or other proceedings in the United States, for example, if there is a court subpoena, unless you have consented for this use.

The Certificate cannot be used to refuse a request for information from personnel of a federal or state agency that is sponsoring the study for auditing or evaluation purposes or for information that must be disclosed in order to meet the requirements of the federal Food and Drug Administration (FDA).

The Certificate of Confidentiality will not be used to prevent disclosure as required by federal, state, or local law, such as mandatory reporting requirements for child abuse or neglect, disabled adult abuse or neglect, communicable diseases, injuries caused by suspected criminal violence, cancer diagnosis or benign brain or central nervous system tumors or other mandatory reporting requirement under applicable law. The Certificate of Confidentiality will not be used if disclosure is for other scientific research, as allowed by federal regulations protecting research subjects or for any purpose you have consented to in this informed consent document.

You should understand that a Certificate of Confidentiality does not prevent you from voluntarily releasing information about yourself or your involvement in this research. If an insurer, employer, or other person obtains your written consent to receive research information, then the researchers may not use the Certificate to withhold that information.

What will happen if you are injured by this research?

All research involves a chance that something bad might happen to you. If you are hurt, become sick, or develop a reaction from something that was done as part of this study, the researcher will help you get medical care. The University of North Carolina at Chapel Hill has not set aside funds to pay you for any such injuries, illnesses or reactions, or for the related medical care. Any costs for medical expenses will be billed to you or your insurance company. You may be responsible for any co-payments and your insurance may not cover the costs of study related injuries.

If you think you have been injured from taking part in this study, call the Principal Investigator at the phone number provided on this consent form. They will let you know what you should do.
By signing this form, you do not give up your right to seek payment or other rights if you are harmed as a result of being in this study.
What if you want to stop before your part in the study is complete?
You can withdraw from this study at any time, without penalty. The investigators also have the right to stop your participation at any time. This could be because you have had an unexpected reaction, or have failed to follow instructions, or because the entire study has been stopped.

If you withdraw or are withdrawn from this study all data collected up until the point of withdrawal will be kept. No new information will be collected unless you provide additional written permission for further data collection at the time of your withdrawal.
Will you receive anything for being in this study?

Participants will receive $20 for completing the 6 and 12 month study sessions ($20 at 6 months and $20 at 12 months). Select participants will receive $20 for completing the 6 and 12 month interviews ($20 at 6 month and $20 at 12 months). If you complete all study sessions, you will receive $40 in total. If you complete both a 6 month and 12 month interview, you will receive $30 in total. If you complete all research visits and 2 interviews, you will receive $70 in total.
Any payment provided for participation in this study may be subject to applicable tax withholding obligations

Will it cost you anything to be in this study?
It will not cost you anything to be in this study.

Who is sponsoring this study?
This research is funded by the National Institutes of Health. This means that the research team is being paid by the sponsor for doing the study. The researchers do not, however, have a direct financial interest with the sponsor or in the results of the study.
Outside of the specific funding for this study, Jean Terranova, one of the study investigators, is an employee of Community Servings. John Buse, another study investigator, has a role in The National Center for Advancing Translational Science (NCATS) Clinical & Transitional Science of Awards Program Steering Committee(CTSA).

What if you have questions about this study?
You have the right to ask, and have answered, any questions you may have about this research. If you have questions about the study (including payments), complaints, concerns, or if a research-related injury occurs, you should contact the researchers listed on the first page of this form.

A description of this clinical trial will be available on www.clinicaltrials.gov, as required by U.S. Law. This website will not include information that can identify you. At most, the website will include a summary of the results. You can search this website at any time.

What if you have questions about your rights as a research participant?
All research on human volunteers is reviewed by a committee called the Institutional Review Board or IRB. This committee works to protect your rights and welfare. If you have questions or concerns about your rights, would like more information or have suggestions please contact them. You can call the Institutional Review Board at 919-966-3113 or you can email IRB_subjects@unc.edu.

Are there any other questions I can answer for you?
{If No, proceed below}

Do you voluntarily agree to participate in this research study?

Yes {Proceed}
No {Say: "That is OK. Thank you for your time.", and end call}

10.1.1.2 CONSENT PROCEDURES AND DOCUMENTATION

We begin by reviewing the referral forms for participants referred to Community Servings to preliminarily assess eligibility (e.g., diagnosis of type 2 diabetes, hemoglobin A1c and body mass index in range).

Then, we contact potentially eligible participants to assess eligibility and interest

- Our call protocol is as follows:
  - We will call up to 5 times over a 2 week period. We will attempt 2 calls during regular work hours (8:30 to 5:30 on weekdays) and 3 after hours, including Mon-Thurs evening between 5:31 and 9:00 pm and on Saturday calling between 8:30 and 5:30 pm.
  - We will leave up to 2 phone messages.
- During eligibly calls, data are entered into the REDCap eligibility questionnaire. This form has text for the phone call.
- If a potential participant is interested and eligible, staff will either mail or email the consent forms to this individual and an appointment will be made for the consent phone call in about 1 week.

Consent phone call

- At this phone call, key components of the consent form are reviewed in detail to ensure that the potential participant understands what participation entails. If the potential participant agrees to participate, they will provide verbal consent, which will be documented in REDCap

10.1.2 STUDY DISCONTINUATION AND CLOSURE

Given the low risk nature of the study, we do not anticipate stopping for adverse outcomes.

The Independent Safety Monitor may stop the study based on their assessment of "futility" attributed to poor participant enrollment.

10.1.3 CONFIDENTIALITY AND PRIVACY

Participant confidentiality and privacy is strictly held in trust by the participating investigators, their staff, the safety and oversight monitor(s), and the sponsor(s) and funding agency. This confidentiality is extended to the data being collected as part of this study. Data that could be used to identify a specific study participant will be held in strict confidence within the research team. No personally-identifiable information from the study will be released to any unauthorized third party without prior written approval of the sponsor/funding agency.

All research activities will be conducted in as private a setting as possible.

The study monitor, other authorized representatives of the sponsor or funding agency, representatives of the Institutional Review Board (IRB), regulatory agencies or representatives from companies or organizations supplying the product, may inspect all documents and records required to be maintained by the investigator, including but not limited to, medical records (office, clinic, or hospital) and pharmacy records for the participants in this study. The clinical study site will permit access to such records.

The study participant's contact information will be securely stored for internal use during the study. At the end of the study, all records will continue to be kept in a secure location for as long a period as dictated by the reviewing IRB, Institutional policies, or sponsor/funding agency requirements.

Study participant research data, which is for purposes of statistical analysis and scientific reporting, will be stored in analytic dataset on secure university servers. This will not include the participant's contact or identifying information. Rather, individual participants and their research data will be identified by a unique study identification number. The study data entry and study management systems used by this study will be secured and password protected. At the end of the study, all study databases will be de-identified and archived.

Measures Taken to Ensure Confidentiality of Data Shared per the NIH Data Sharing Policies

It is NIH policy that the results and accomplishments of the activities that it funds should be made available to the public (see *https://grants.nih.gov/policy/sharing.htm*). The PI will ensure all mechanisms used to share data will include proper plans and safeguards for the protection of privacy, confidentiality, and security for data dissemination and reuse (e.g., all data will be thoroughly de-identified and will not

be traceable to a specific study participant). Plans for archiving and long-term preservation of the data will be implemented, as appropriate.

Certificate of Confidentiality

To further protect the privacy of study participants, the Secretary, Health and Human Services (HHS), has issued a Certificate of Confidentiality (CoC) to all researchers engaged in biomedical, behavioral, clinical or other human subjects research funded wholly or in part by the federal government. Recipients of NIH funding for human subjects research are required to protect identifiable research information from forced disclosure per the terms of the NIH Policy (see *https://humansubjects.nih.gov/coc/index*). As set forth in 45 CFR Part 75.303(a) and NIHGPS Chapter 8.3, recipients conducting NIH-supported research covered by this Policy are required to establish and maintain effective internal controls (e.g., policies and procedures) that provide reasonable assurance that the award is managed in compliance with Federal statutes, regulations, and the terms and conditions of award. It is the NIH policy that investigators and others who have access to research records will not disclose identifying information except when the participant consents or in certain instances when federal, state, or local law or regulation requires disclosure. NIH expects investigators to inform research participants of the protections and the limits to protections provided by a Certificate issued by this Policy.

10.1.4 FUTURE USE OF STORED SPECIMENS AND DATA

- No specimens will be stored
- Data will be stored on secured servers administered by UNC-Chapel Hill IT.
- At the time of this protocol preparation, data will be stored indefinitely. This is subject to change owing to regulatory requirements.

10.1.5 KEY ROLES AND STUDY GOVERNANCE

The name and contact information of the Principal Investigator and the Independent Safety Monitor are listed below.

Principal Investigator	**Independent Safety Monitor**
Seth A. Berkowitz, MD MPH *Assistant Professor of Medicine* *Division of General Medicine and Clinical Epidemiology* *Department of Medicine* *School of Medicine*	*Jacqueline Halladay, MD MPH* *Professor of Medicine* *Chair of Research, UNC Health Sciences at MAHEC* *Department of Family Medicine* *Department of Medicine* *School of Medicine*

UNC-Chapel Hill	*UNC-Chapel Hill*
5034 Old Clinic Building *CB #7110* *Chapel Hill, NC 27599*	*Mountain Area Health Education Center* *121 Hendersonville Road* *Asheville, NC 28803-2868*
919-966-2276	*919-843-6607*
seth_berkowitz@med.unc.edu	*jacqueline_halladay@med.unc.edu*

This single site study has 2 primary committees

- Operations Committee: Includes site-PIs, project managers, research dietitians, and other investigators and will RAs as appropriate. Meets weekly.
- Co-I Committee: Includes co-investigators and meets as appropriate to discuss major study protocol issues and later to review and interpret the data.

Oversight Committee

- Data Safety Monitoring Committee: Meets every 6 months.

10.1.6 SAFETY OVERSIGHT

Safety oversite is provided by a data safety monitoring committee. The charter, dated March 26, 2021 is below:

Data Safety Monitoring Committee with Independent Safety Monitor Charter

March 26, 2021

The investigators believe the interventions and measurement protocols pose minimal risk to participants. Because of this study's low risk status, the data safety monitoring plan focuses on careful monitoring of the study's progress as well as adverse events.

This study will use an Independent Safety Monitoring Plan wherein the principal investigator, independent safety monitor, and study biostatistician will comprise the data safety monitoring committee (DSMC). The study project manager will provide administrative assistance to the committee.

- Principal Investigator: Seth A. Berkowitz, MD MPH
- Independent Safety Monitor: Jacqueline Halladay, MD MPH
- Study Biostatistician: Paul Stewart, PhD
- Project Manager: Katharine Ball Ricks, PhD

Key personnel involved with the logistics of participant enrollment, participation, and follow-up will meet regularly (usually weekly) to review, among other things, progress on accrual, follow-up, and adverse events. They will then prepare the following reports for the DSMC:

Data Type	Frequency of Review
Participant accrual	Weekly, with **monthly** summary reports for the principal investigator and quarterly reports to the independent safety monitor.
Adverse events	These will be reported as they occur to the IRB with **monthly** reports to the principal investigator. The monthly reports will be combined into quarterly reports for the independent safety monitor. Any potentially study related death or severe and unexpected adverse event will be reported to the principal investigator within 1 working day. If the event is determined to be study related, it will be reported to the independent safety monitor within 1 week.
Adherence with treatment	For the medically tailored meal group, study process measures will include meals delivered and sessions attended. For the food subsidy group, process measures will include redemption of the food subsidy. Adherence reports will be prepared for **monthly** reports to the principal investigator.
Follow-up rates for data collection (including the number of participants eligible for and completing follow-up)	Weekly, with **monthly** summary reports prepared for the principal investigator.
Stopping rules with regard to benefit	Not applicable
Stopping rules with regard to statistical power	The principal investigator and the study biostatistician will provide annual review.
Stopping rules with regard to adverse events	The Independent Safety Monitor will received quarterly reports of adverse events as above. The Independent Safety Monitor will provide an annual review letter including their determination as to whether there is a signal of risk or not.

Operational definition of futility with regard to recruitment: Less than one half of the anticipated sample size enrolled during the enrollment period. This may be modified by the voting members of the DSMC.

Meetings and responsibilities of the DSMC: Prior to enrollment, all DSMC will review all protocols that may have a bearing on participant accrual or safety. The DSMC will subsequently meet every 6 months until completion of the study. In addition, any committee member may convene additional meetings as needed. The primary role of the DSMC will be to monitor study data as outlined in the table above. The

committee will also assist with any necessary modifications of protocols to address unexpected problems with subject accrual, subject attrition, or adverse events.

Measurement and reporting of adverse events (AEs): The study team will monitor adverse events that include, but are not limited to the following: deaths, hospitalizations, emergency room visits, and hypoglycemia (defined as blood glucose < 60 mg/d with symptoms). We will specifically inquire about these adverse events at all follow-up measurement visits and during check-in calls. The PI will be responsible for completing an AE report when a reportable adverse event is recognized and will submit this report to the UNC IRB. The UNC IRB definitions, standards, and forms will be used for reporting AEs. Also, at measurement visits, we will inquire about angina, myocardial infarction, and stroke.

Assessment of Adverse Events: The principal investigator and independent safety monitor will be blinded to study arms when assessing adverse events. The project manager and study biostatistician may be unblinded as needed to provide necessary information to assess risks to participants.

10.1.7 CLINICAL MONITORING

N/A.

10.1.8 QUALITY ASSURANCE AND QUALITY CONTROL

Data Capture: We will use Research Electronic Data Capture or REDCap as our primary application for data capture in this study. REDCap is a secure web application use to build and manage surveys and other data capture mechanisms for clinical research. A primary reason for using REDCap is that this is the research data management service support by NC TraCS (UNC's CTSA). Below is information in support of REDCap use at UNC from NC TraCS.

"The TraCS Clinical Research Data Management Service is a key initiative of the Biomedical Informatics core of the UNC-Chapel Hill CTSA. The purpose is to provide a system, and associated support resources, to enable efficient and high quality collection and management of research data that is standards-based in design, development and implementation.

Standard features of electronic clinical research data management systems are available in the web-based systems provided with the service. These include interactive data entry with real-time field validation, lab data imports, audit logs to record database modifications, database integrity checks, security (in logins, permissions based on need, and encryption), reporting, forms inventory, and exports to common statistical packages for analysis.

Logging tracks all data entered in REDCap so that it can be traced back to the person who entered it. No data can be changed without showing who has made the changes. This allows the study team to ensure the security and integrity of the data collected and submitted; therefore, there are controls surrounding this aspect.

REDCap also provides for principle investigator sign-off on data, as required in FDA studies. Although users can modify data based on their permissions, they cannot delete the subject or history of that subject. Requests to delete a subject must be made to the REDCap system administrator.

Our database system provides for secure web-based data entry with the data stored on servers that we maintain. The data is encrypted during transmission. The servers are located in a secure campus area with all appropriate physical security measures in place. The web and database servers are monitored by the TraCS IT staff, patched frequently, and scanned by a third party vendor to ensure that they are protected against known vulnerabilities. The scanning application is the standard service for the entire campus. Access is by individual user id, and is restricted to the forms and/or functions that the user needs to have.

The applications themselves are written using open source tools, and have also been scanned by the campus security office to ensure that the applications are also protected from known exploits. The data is backed up to electronic media on a daily basis. The electronic media is secured by ITS stored in a secure area separate from the servers."

REDCap will be used for survey and field data collection. Lab data will be imported into REDCap. The REDCap data will reside on a university server.

Table outlining data sources:

Data Source	Who will Collect the Data	Data Storage
Screening data	Research staff will collect data on: --Those eligible, who agree to participate are assigned a study ID # and identifiable data is collected as outlined below, after they sign the study consent form. --Those eligible, who decline participation, are assigned a non-participant ID #, and limited de-identified data will be stored.	University server for this project, hereafter referred to as university server
Questionnaire data	Research staff	University server
Physiologic data: weight, blood pressure	Research staff	University server
HbA1c	Specimens collected by trained phlebotomists. Specimens analyzed at LabCorp lab certified	University server

	by NGSP for HbA1c results (traceable to assay for DCCT). Results sent securely to research staff for data entry.	
Taped recordings of intervention counseling sessions	Research staff will record the session and upload digital files.	University server
Qualitative interview sessions	Research staff will record the interview and upload digital files.	University server

Data Management: A study analyst based at the Cecil G. Sheps Center for Health Services Research (Sheps Center) will provide programming related to database management. This includes conversion of data into SAS datasets, data editing, and creation of code books for each data collection time point and for the study's process data. These data sets will be used by the statistical co-investigators.

To ensure high quality data, in addition to range checks embedded in REDCap and features to support data quality outlined above, the study project manager will review study data on a weekly basis. He/she will investigate any data irregularities including missing data. Monthly, he/she will meet with the study PI to review study data management. Any inconsistencies or concerns will lead to data queries to be resolved by the study analyst.

Monitoring for Quality and Consistency of the Intervention: Fidelity of intervention delivery will be assessed using direct observation and audio recordings of up to 20% of sessions by Co-investigator Ms. Delahanty.

10.1.9 DATA HANDLING AND RECORD KEEPING

10.1.9.1 DATA COLLECTION AND MANAGEMENT RESPONSIBILITIES

The data issues to be described in this section are covered in Section 10.1.8.

10.1.9.2 STUDY RECORDS RETENTION

At time of writing this protocol, storage of study data are indefinite. This is subject to change owing to regulatory requirements.

10.1.10 PROTOCOL DEVIATIONS

This study will follow the protocol deviation guidelines from the UNC IRB, as outlined below:

Purpose: To record all protocol deviations that occur at a study site for both observational and interventional clinical research studies.
IMPORTANT: This log is maintained in the Study Binder (Synonyms for this binder include Investigator Binder, Regulatory Binder, Investigator Site File [ISF], and Study File.) and should be made available upon request for review by the IRB and the Sponsor's monitor. Deviations should be reported to the IRB of record as per the IRB Standard Operating Procedures. See OHRE/IRB SOP 1401 for reporting requirements for deviations to the UNC IRB.

Audience/User: Study coordinators, principal investigator (PI), other site staff, clinical monitor

Best Practice Recommendations:

- Record protocol deviations in the tracking log as they occur, to ensure completeness and accuracy of the data.
- The site PI should sign each form after it has been completed or immediately prior to a monitoring visit. If it has been signed with fewer than five deviations entered into it, the next identified deviation should be reported on a new page to ensure that all deviations have been reviewed by the PI.
- Number each page and identify the final page of the log by indicating FINAL in the page number field.
- Store pages in reverse chronological order, with the newest pages of the log placed at the front of the section.
- Remove this page before using the log.

Example Deviation Form

***DEVIATION CATEGORIES:**

A. Informed Consent
B. Eligibility
C. Protocol implementation
D. Reporting
E. Other, specify in log

**DEVIATION CODES: Numbers listed by the sample protocol deviations

Informed Consent (Category A)

1. Failure to obtain informed consent
2. Consent form used was not current IRB-approved version
3. Consent form does not include updates or information required by IRB
4. Consent form missing
5. Consent form not signed and dated by participant
6. Consent form does not contain all required signatures
7. Other, specify in log

Eligibility (Category B)

8. Participant did not meet eligibility criterion
9. Randomization of an ineligible participant
10. Participant randomized prior to completing Baseline Assessment, etc.
11. Randomization and/or treatment of participant prior to IRB approval of protocol
12. Other, specify in log

Protocol implementation (Category C)

13. Failure to keep IRB approval up to date
14. Participant receives wrong treatment
15. Participant seen outside visit window
16. Use of unallowable concomitant treatments
17. Prescribed dosing outside protocol guidelines
18. Missed assessment
19. Laboratory tests not done
20. Missed visit
21. Other, specify in log

Reporting (Category D)

22. Not submitting reportable information to the IRB within 7 days
23. Failure to respond to the NSI stipulations in the requested timeframe
24. Other, specify in log

Other (Category E)

25. Other, specify in log

IRB Study #					Site Name/Number:				
Protocol Title (Abbreviated):					Protocol ID/Number:				
Principal Investigator:					Page number [1]:				
Ref No.	Subject ID	Date of Deviation	Date Identified	Deviation Description	Dev. Type [2]	Resulted in AE?	Did Subject Continue in Study?	Meets IRB Reporting Req. (I.e. NSI) (Yes/No)	IRB Reporting Date
1									
2									
3									
4									
5									
6									
7									

10.1.11 PUBLICATION AND DATA SHARING POLICY

This study will be conducted in accordance with the following publication and data sharing policies and regulations:

National Institutes of Health (NIH) Public Access Policy, which ensures that the public has access to the published results of NIH funded research. It requires scientists to submit final peer-reviewed journal manuscripts that arise from NIH funds to the digital archive PubMed Central upon acceptance for publication.

This study will comply with the NIH Data Sharing Policy and Policy on the Dissemination of NIH-Funded Clinical Trial Information and the Clinical Trials Registration and Results Information Submission rule. As such, this trial will be registered at ClinicalTrials.gov, and results information from this trial will be submitted to ClinicalTrials.gov. In addition, every attempt will be made to publish results in peer-reviewed journals. Data from this study may be requested from other researchers 5 years after the completion of the study by contacting the principle investigator. Considerations for ensuring confidentiality of these shared data are described in Section 10.1.3.

In addition, this study will comply with the NIH Genomic Data Sharing Policy, which applies to all NIH-funded research that generates large-scale human or non-human genomic data, as well as the use of these data for subsequent research. Large-scale data include genome-wide association studies (GWAS), single nucleotide polymorphisms (SNP) arrays, and genome sequence, transcriptomic, epigenomic, and gene expression data.

10.1.12 CONFLICT OF INTEREST POLICY

The independence of this study from any actual or perceived influence, such as by the pharmaceutical industry, is critical. Therefore, any actual conflict of interest of persons who have a role in the design, conduct, analysis, publication, or any aspect of this trial will be disclosed and managed. Furthermore, persons who have a perceived conflict of interest will be required to have such conflicts managed in a way that is appropriate to their participation in the design and conduct of this trial according to the guidelines set forth by the Conflict of Interest Program at UNC.

10.2 ADDITIONAL CONSIDERATIONS

N/A.

10.3 ABBREVIATIONS AND SPECIAL TERMS

AE	Adverse Event
ANCOVA	Analysis of Covariance
CFR	Code of Federal Regulations
CLIA	Clinical Laboratory Improvement Amendments
CMP	Clinical Monitoring Plan
COC	Certificate of Confidentiality
CONSORT	Consolidated Standards of Reporting Trials
CRF	Case Report Form
DCC	Data Coordinating Center
DHHS	Department of Health and Human Services
DSMB	Data Safety Monitoring Board
DRE	Disease-Related Event
EC	Ethics Committee
eCRF	Electronic Case Report Forms
FDA	Food and Drug Administration
FDAAA	Food and Drug Administration Amendments Act of 2007
FFR	Federal Financial Report
GCP	Good Clinical Practice
GLP	Good Laboratory Practices
GMP	Good Manufacturing Practices
GWAS	Genome-Wide Association Studies
HIPAA	Health Insurance Portability and Accountability Act
IB	Investigator's Brochure
ICH	International Council on Harmonisation
ICMJE	International Committee of Medical Journal Editors
IDE	Investigational Device Exemption
IND	Investigational New Drug Application
IRB	Institutional Review Board
ISM	Independent Safety Monitor
ITT	Intention-To-Treat
LSMEANS	Least-squares Means
MedDRA	Medical Dictionary for Regulatory Activities
MOP	Manual of Procedures
NCT	National Clinical Trial
NIH	National Institutes of Health
NIH IC	NIH Institute or Center
OHRP	Office for Human Research Protections
PI	Principal Investigator
QA	Quality Assurance
QC	Quality Control
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan

SMC	Safety Monitoring Committee
SOA	Schedule of Activities
SOC	System Organ Class
SOP	Standard Operating Procedure
UP	Unanticipated Problem
US	United States

10.4 PROTOCOL AMENDMENT HISTORY

*The table below is intended to capture changes of IRB-approved versions of the protocol, including a description of the change and rationale. A **Summary of Changes** table for the current amendment is located in the **Protocol Title Page**.*

Version	Date	Description of Change	Brief Rationale
1.0	4/29/21	Initial Protocol Document	This serves as the current protocol immediately prior to the enrollment of the first participant
1.1	6/1/21	Added PAID-11 to measures section	This measure had been inadvertently omitted in the protocol document, but had been a planned measure, was already IRB approved, and was programmed into REDCaP questionnaire. This change occurred prior to enrollment of first participant.
1.2	3/2/22	Changed qualitative interview reimbursement from $15 to $20	Make reimbursement for qualitative interview equal to reimbursement for study visits
1.3	9/21/22	Updated missing data strategy to include use of measures obtained in routine care	Using measures obtained in routine care during the study period will help address missing data (e.g., if a participant does not have a 6-month study measure of HbA1c, a measure from their primary care clinic might provide additional useful information)
1.4	12/6/22	Change in food subsidy from $30 to $40, addition of 3-month incentive, increase in 12 month visit incentive to $50 from $20	The U.S. experienced a higher than typical period of inflation during the study period, making the relative value of the initial $30/food subsidy decrease relative to MTM. We also added additional check-points and adjusted visit incentive amounts to help with

			recruitment and retention of participants.
1.41	7/25/23	Clarification that exclusion criterion related to prior bariatric surgery applies to not having had bariatric surgery in the prior 2 years Clarification in data collection processes to allow data from electronic health records to fill if study visit data missing	Clarification on length of time related to bariatric surgery. Purpose is to help minimize other sources of weight fluctuation during the study period. To supplement data collection from VNA visits, we will allow measurements from the electronic health record to be used. We have chosen to adopt these methods to increase the precision of the study estimates. At the time this decision was made, data collection is still ongoing and no data have been analyzed.

11 REFERENCES

1. USDA ERS - Food Security in the U.S. [Internet]. [cited 2019 Jul 11]. Available from: https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/

2. USDA. Food Security Status of U.S. Households in 2018 [Internet]. [cited 2019 Sep 18]. Available from: https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/key-statistics-graphics.aspx#foodsecure

3. Berkowitz SA, Baggett TP, Wexler DJ, Huskey KW, Wee CC. Food insecurity and metabolic control among U.S. adults with diabetes. Diabetes Care. 2013 Oct;36(10):3093–9. doi:10.2337/dc13-0570 PubMed PMID: 23757436; PubMed Central PMCID: PMC3781549.

4. Berkowitz SA, Berkowitz TSZ, Meigs JB, Wexler DJ. Trends in food insecurity for adults with cardiometabolic disease in the United States: 2005-2012. PloS One. 2017;12(6):e0179172. doi:10.1371/journal.pone.0179172 PubMed PMID: 28591225; PubMed Central PMCID: PMC5462405.

5. Seligman HK, Laraia BA, Kushel MB. Food insecurity is associated with chronic disease among low-income NHANES participants. J Nutr. 2010 Feb;140(2):304–10. doi:10.3945/jn.109.112573 PubMed PMID: 20032485; PubMed Central PMCID: PMC2806885.

6. Seligman HK, Bindman AB, Vittinghoff E, Kanaya AM, Kushel MB. Food insecurity is associated with diabetes mellitus: results from the National Health Examination and Nutrition Examination Survey (NHANES) 1999-2002. J Gen Intern Med. 2007 Jul;22(7):1018–23. doi:10.1007/s11606-007-0192-6 PubMed PMID: 17436030; PubMed Central PMCID: PMC2583797.

7. Orr CJ, Keyserling TC, Ammerman AS, Berkowitz SA. Diet quality trends among adults with diabetes by socioeconomic status in the U.S.: 1999-2014. BMC Endocr Disord. 2019 May 31;19(1):54. doi:10.1186/s12902-019-0382-3 PubMed PMID: 31151439; PubMed Central PMCID: PMC6544994.

8. Centers for Disease Control and Prevention. National Diabetes Statistics Report, 2017 [Internet]. Centers for Disease Control and Prevention, U.S. Dept of Health and Human Services; 2017 [cited 2017 Aug 10]. Available from: https://www.cdc.gov/diabetes/pdfs/data/statistics/national-diabetes-statistics-report.pdf

9. Coleman-Jensen A, Rabbitt MP, Gregory CA, Singh A. Household Food Security in the United States in 2017 [Internet]. [cited 2018 Sep 25]. Available from: https://www.ers.usda.gov/publications/pub-details/?pubid=90022

10. Berkowitz SA, Karter AJ, Corbie-Smith G, Seligman HK, Ackroyd SA, Barnard LS, et al. Food Insecurity, Food “Deserts,” and Glycemic Control in Patients With Diabetes: A Longitudinal

Analysis. Diabetes Care. 2018 Mar 19;dc171981. doi:10.2337/dc17-1981 PubMed PMID: 29555650.

11. Berkowitz SA, Meigs JB, DeWalt D, Seligman HK, Barnard LS, Bright OJM, et al. Material need insecurities, control of diabetes mellitus, and use of health care resources: results of the Measuring Economic Insecurity in Diabetes study. JAMA Intern Med. 2015 Feb;175(2):257–65. doi:10.1001/jamainternmed.2014.6888 PubMed PMID: 25545780; PubMed Central PMCID: PMC4484589.

12. Mayer VL, McDonough K, Seligman H, Mitra N, Long JA. Food insecurity, coping strategies and glucose control in low-income patients with diabetes. Public Health Nutr. 2016 Apr;19(6):1103–11. doi:10.1017/S1368980015002323 PubMed PMID: 26328922.

13. Shalowitz MU, Eng JS, McKinney CO, Krohn J, Lapin B, Wang CH, et al. Food security is related to adult type 2 diabetes control over time in a United States safety net primary care clinic population. Nutr Diabetes. 2017 May 15;7(5):e277. doi:10.1038/nutd.2017.18 PubMed PMID: 28504709; PubMed Central PMCID: PMC5518800.

14. Walker RJ, Gebregziabher M, Martin-Harris B, Egede LE. Quantifying direct effects of social determinants of health on glycemic control in adults with type 2 diabetes. Diabetes Technol Ther. 2015 Feb;17(2):80–7. doi:10.1089/dia.2014.0166 PubMed PMID: 25361382; PubMed Central PMCID: PMC4322090.

15. Walker RJ, Strom Williams J, Egede LE. Influence of Race, Ethnicity and Social Determinants of Health on Diabetes Outcomes. Am J Med Sci. 2016 Apr;351(4):366–73. doi:10.1016/j.amjms.2016.01.008 PubMed PMID: 27079342; PubMed Central PMCID: PMC4834895.

16. Walker RJ, Williams JS, Egede LE. Pathways between food insecurity and glycaemic control in individuals with type 2 diabetes. Public Health Nutr. 2018;21(17):3237–44. doi:10.1017/S1368980018001908 PubMed PMID: 30088467; PubMed Central PMCID: PMC6258341.

17. Smalls BL, Gregory CM, Zoller JS, Egede LE. Assessing the relationship between neighborhood factors and diabetes related health outcomes and self-care behaviors. BMC Health Serv Res. 2015 Oct 1;15:445. doi:10.1186/s12913-015-1086-7 PubMed PMID: 26428459; PubMed Central PMCID: PMC4589943.

18. Seligman HK, Jacobs EA, López A, Tschann J, Fernandez A. Food insecurity and glycemic control among low-income patients with type 2 diabetes. Diabetes Care. 2012 Feb;35(2):233–8. doi:10.2337/dc11-1627 PubMed PMID: 22210570; PubMed Central PMCID: PMC3263865.

19. Heerman WJ, Wallston KA, Osborn CY, Bian A, Schlundt DG, Barto SD, et al. Food insecurity is associated with diabetes self-care behaviours and glycaemic control. Diabet Med J Br Diabet

Assoc. 2016;33(6):844–50. doi:10.1111/dme.12896 PubMed PMID: 26314941; PubMed Central PMCID: PMC4769979.

20. Garcia SP, Haddix A, Barnett K. Incremental Health Care Costs Associated With Food Insecurity and Chronic Conditions Among Older Adults. Prev Chronic Dis. 2018 Aug 30;15:E108. doi:10.5888/pcd15.180058 PubMed PMID: 30171678.

21. Crews DC, Kuczmarski MF, Grubbs V, Hedgeman E, Shahinian VB, Evans MK, et al. Effect of food insecurity on chronic kidney disease in lower-income Americans. Am J Nephrol. 2014;39(1):27–35. doi:10.1159/000357595 PubMed PMID: 24434743; PubMed Central PMCID: PMC3952065.

22. Gibson DM. Food Insecurity, Eye Care Receipt, and Diabetic Retinopathy Among US Adults with Diabetes: Implications for Primary Care. J Gen Intern Med. 2019 Apr 10. doi:10.1007/s11606-019-04992-x PubMed PMID: 30972552.

23. Redmond ML, Dong F, Goetz J, Jacobson LT, Collins TC. Food Insecurity and Peripheral Arterial Disease in Older Adult Populations. J Nutr Health Aging. 2016;20(10):989–95. doi:10.1007/s12603-015-0639-0 PubMed PMID: 27925138.

24. Silverman J, Krieger J, Kiefer M, Hebert P, Robinson J, Nelson K. The Relationship Between Food Insecurity and Depression, Diabetes Distress and Medication Adherence Among Low-Income Patients with Poorly-Controlled Diabetes. J Gen Intern Med. 2015 Oct;30(10):1476–80. doi:10.1007/s11606-015-3351-1 PubMed PMID: 25917659; PubMed Central PMCID: PMC4579205.

25. Lyles CR, Wolf MS, Schillinger D, Davis TC, Dewalt D, Dahlke AR, et al. Food insecurity in relation to changes in hemoglobin A1c, self-efficacy, and fruit/vegetable intake during a diabetes educational intervention. Diabetes Care. 2013 Jun;36(6):1448–53. doi:10.2337/dc12-1961 PubMed PMID: 23275354; PubMed Central PMCID: PMC3661820.

26. Vijayaraghavan M, Jacobs EA, Seligman H, Fernandez A. The association between housing instability, food insecurity, and diabetes self-efficacy in low-income adults. J Health Care Poor Underserved. 2011 Nov;22(4):1279–91. doi:10.1353/hpu.2011.0131 PubMed PMID: 22080709.

27. Walker RJ, Chawla A, Garacci E, Williams JS, Mendez C, Ozieh MN, et al. Assessing the relationship between food insecurity and mortality among U.S. adults. Ann Epidemiol. 2019 Apr;32:43–8. doi:10.1016/j.annepidem.2019.01.014 PubMed PMID: 30799207; PubMed Central PMCID: PMC6441375.

28. Gundersen C, Tarasuk V, Cheng J, de Oliveira C, Kurdyak P. Food insecurity status and mortality among adults in Ontario, Canada. PloS One. 2018;13(8):e0202642. doi:10.1371/journal.pone.0202642 PubMed PMID: 30138369; PubMed Central PMCID: PMC6133286.

29. Morales ME, Berkowitz SA. The Relationship Between Food Insecurity, Dietary Patterns, and Obesity. Curr Nutr Rep. 2016 Mar 1;5(1):54–60. doi:10.1007/s13668-016-0153-y

30. Berkowitz SA, Gao X, Tucker KL. Food-insecure dietary patterns are associated with poor longitudinal glycemic control in diabetes: results from the Boston Puerto Rican Health study. Diabetes Care. 2014 Sep;37(9):2587–92. doi:10.2337/dc14-0753 PubMed PMID: 24969578; PubMed Central PMCID: PMC4140162.

31. Lee AM, Scharf RJ, DeBoer MD. Food insecurity is associated with prediabetes and dietary differences in U.S. adults aged 20-39. Prev Med. 2018 Nov;116:180–5. doi:10.1016/j.ypmed.2018.09.012 PubMed PMID: 30267733.

32. Leung CW, Tester JM. The Association between Food Insecurity and Diet Quality Varies by Race/Ethnicity: An Analysis of National Health and Nutrition Examination Survey 2011-2014 Results. J Acad Nutr Diet. 2018 Dec 19. doi:10.1016/j.jand.2018.10.011 PubMed PMID: 30579633; PubMed Central PMCID: PMC6584552.

33. Leung CW, Epel ES, Ritchie LD, Crawford PB, Laraia BA. Food insecurity is inversely associated with diet quality of lower-income adults. J Acad Nutr Diet. 2014 Dec;114(12):1943-1953.e2. doi:10.1016/j.jand.2014.06.353 PubMed PMID: 25091796.

34. Seligman HK, Schillinger D. Hunger and socioeconomic disparities in chronic disease. N Engl J Med. 2010 Jul 1;363(1):6–9. doi:10.1056/NEJMp1000072 PubMed PMID: 20592297.

35. Ammerman AS, Hartman T, DeMarco MM. Behavioral Economics and the Supplemental Nutrition Assistance Program:: Making the Healthy Choice the Easy Choice. Am J Prev Med. 2017 Feb 1;52(2):S145–50. doi:10.1016/j.amepre.2016.08.017 PubMed PMID: 28109416.

36. Zellner DA, Loaiza S, Gonzalez Z, Pita J, Morales J, Pecora D, et al. Food selection changes under stress. Physiol Behav. 2006 Apr 15;87(4):789–93. doi:10.1016/j.physbeh.2006.01.014 PubMed PMID: 16519909.

37. Zellner DA, Saito S, Gonzalez J. The effect of stress on men's food selection. Appetite. 2007 Nov;49(3):696–9. doi:10.1016/j.appet.2007.06.013 PubMed PMID: 17826866.

38. Epel E, Lapidus R, McEwen B, Brownell K. Stress may add bite to appetite in women: a laboratory study of stress-induced cortisol and eating behavior. Psychoneuroendocrinology. 2001 Jan;26(1):37–49. PubMed PMID: 11070333.

39. Oliver G, Wardle J, Gibson EL. Stress and food choice: a laboratory study. Psychosom Med. 2000 Dec;62(6):853–65. PubMed PMID: 11139006.

40. Wallis DJ, Hetherington MM. Stress and eating: the effects of ego-threat and cognitive demand on food intake in restrained and emotional eaters. Appetite. 2004 Aug;43(1):39–46. doi:10.1016/j.appet.2004.02.001 PubMed PMID: 15262016.

41. Wallis DJ, Hetherington MM. Emotions and eating. Self-reported and experimentally induced changes in food intake under stress. Appetite. 2009 Apr;52(2):355–62. doi:10.1016/j.appet.2008.11.007 PubMed PMID: 19071171.

42. Jääskeläinen A, Nevanperä N, Remes J, Rahkonen F, Järvelin MR, Laitinen J. Stress-related eating, obesity and associated behavioural traits in adolescents: a prospective population-based cohort study. BMC Public Health. 2014 Apr 7;14:321. doi:10.1186/1471-2458-14-321 PubMed PMID: 24708823; PubMed Central PMCID: PMC3995503.

43. Laugero KD, Falcon LM, Tucker KL. Relationship between perceived stress and dietary and activity patterns in older adults participating in the Boston Puerto Rican Health Study. Appetite. 2011 Feb;56(1):194–204. doi:10.1016/j.appet.2010.11.001 PubMed PMID: 21070827; PubMed Central PMCID: PMC5012221.

44. Seligman HK, Berkowitz SA. Aligning Programs and Policies to Support Food Security and Public Health Goals in the United States. Annu Rev Public Health. 2019 Apr 1;40:319–37. doi:10.1146/annurev-publhealth-040218-044132 PubMed PMID: 30444684.

45. Bowen S, Elliott S, Brenton J. The Joy of Cooking? Contexts. 2014 Aug 1;13(3):20–5. doi:10.1177/1536504214545755

46. Cheung HC, Shen A, Oo S, Tilahun H, Cohen MJ, Berkowitz SA. Food Insecurity and Body Mass Index: A Longitudinal Mixed Methods Study, Chelsea, Massachusetts, 2009-2013. Prev Chronic Dis. 2015 Aug 6;12:E125. doi:10.5888/pcd12.150001 PubMed PMID: 26247425; PubMed Central PMCID: PMC4565511.

47. Basu S, Berkowitz SA, Seligman H. The Monthly Cycle of Hypoglycemia: An Observational Claims-based Study of Emergency Room Visits, Hospital Admissions, and Costs in a Commercially Insured Population. Med Care. 2017 Jul;55(7):639–45. doi:10.1097/MLR.0000000000000728 PubMed PMID: 28481762.

48. Seligman HK, Davis TC, Schillinger D, Wolf MS. Food insecurity is associated with hypoglycemia and poor diabetes self-management in a low-income sample with diabetes. J Health Care Poor Underserved. 2010 Nov;21(4):1227–33. doi:10.1353/hpu.2010.0921 PubMed PMID: 21099074; PubMed Central PMCID: PMC3004431.

49. Seligman HK, Jacobs EA, Lopez A, Sarkar U, Tschann J, Fernandez A. Food insecurity and hypoglycemia among safety net patients with diabetes. Arch Intern Med. 2011 Jul 11;171(13):1204–6. doi:10.1001/archinternmed.2011.287 PubMed PMID: 21747017; PubMed Central PMCID: PMC4230711.

50. Seligman HK, Bolger AF, Guzman D, López A, Bibbins-Domingo K. Exhaustion of food budgets at month's end and hospital admissions for hypoglycemia. Health Aff Proj Hope. 2014

Jan;33(1):116–23. doi:10.1377/hlthaff.2013.0096 PubMed PMID: 24395943; PubMed Central PMCID: PMC4215698.

51. Berkowitz SA, Seligman HK, Choudhry NK. Treat or eat: food insecurity, cost-related medication underuse, and unmet needs. Am J Med. 2014 Apr;127(4):303-310.e3. doi:10.1016/j.amjmed.2014.01.002 PubMed PMID: 24440543.

52. Herman D, Afulani P, Coleman-Jensen A, Harrison GG. Food Insecurity and Cost-Related Medication Underuse Among Nonelderly Adults in a Nationally Representative Sample. Am J Public Health. 2015 Oct;105(10):e48-59. doi:10.2105/AJPH.2015.302712 PubMed PMID: 26270308; PubMed Central PMCID: PMC4566547.

53. Afulani P, Herman D, Coleman-Jensen A, Harrison GG. Food Insecurity and Health Outcomes Among Older Adults: The Role of Cost-Related Medication Underuse. J Nutr Gerontol Geriatr. 2015;34(3):319–42. doi:10.1080/21551197.2015.1054575 PubMed PMID: 26267444.

54. Ippolito MM, Lyles CR, Prendergast K, Marshall MB, Waxman E, Seligman HK. Food insecurity and diabetes self-management among food pantry clients. Public Health Nutr. 2017 Jan;20(1):183–9. doi:10.1017/S1368980016001786 PubMed PMID: 27406399.

55. Sattler ELP, Lee JS. Persistent food insecurity is associated with higher levels of cost-related medication nonadherence in low-income older adults. J Nutr Gerontol Geriatr. 2013;32(1):41–58. doi:10.1080/21551197.2012.722888 PubMed PMID: 23451845.

56. Patel MR, Piette JD, Resnicow K, Kowalski-Dobson T, Heisler M. Social Determinants of Health, Cost-related Nonadherence, and Cost-reducing Behaviors Among Adults With Diabetes: Findings From the National Health Interview Survey. Med Care. 2016;54(8):796–803. doi:10.1097/MLR.0000000000000565 PubMed PMID: 27219636; PubMed Central PMCID: PMC4945373.

57. Knight CK, Probst JC, Liese AD, Sercye E, Jones SJ. Household food insecurity and medication "scrimping" among US adults with diabetes. Prev Med. 2016 Feb;83:41–5. doi:10.1016/j.ypmed.2015.11.031 PubMed PMID: 26656403.

58. Gundersen C, Ziliak JP. Food Insecurity And Health Outcomes. Health Aff Proj Hope. 2015 Nov;34(11):1830–9. doi:10.1377/hlthaff.2015.0645 PubMed PMID: 26526240.

59. Barnard LS, Wexler DJ, DeWalt D, Berkowitz SA. Material need support interventions for diabetes prevention and control: a systematic review. Curr Diab Rep. 2015 Feb;15(2):574. doi:10.1007/s11892-014-0574-1 PubMed PMID: 25620406.

60. Long JA, Jahnle EC, Richardson DM, Loewenstein G, Volpp KG. Peer mentoring and financial incentives to improve glucose control in African American veterans: a randomized trial. Ann

Intern Med. 2012 Mar 20;156(6):416–24. doi:10.7326/0003-4819-156-6-201203200-00004 PubMed PMID: 22431674; PubMed Central PMCID: PMC3475415.

61. Heisler M, Vijan S, Makki F, Piette JD. Diabetes control with reciprocal peer support versus nurse care management: a randomized trial. Ann Intern Med. 2010 Oct 19;153(8):507–15. doi:10.7326/0003-4819-153-8-201010190-00007 PubMed PMID: 20956707; PubMed Central PMCID: PMC4117390.

62. Kangovi S, Mitra N, Norton L, Harte R, Zhao X, Carter T, et al. Effect of Community Health Worker Support on Clinical Outcomes of Low-Income Patients Across Primary Care Facilities: A Randomized Clinical Trial. JAMA Intern Med. 2018 01;178(12):1635–43. doi:10.1001/jamainternmed.2018.4630 PubMed PMID: 30422224; PubMed Central PMCID: PMC6469661.

63. Kangovi S, Mitra N, Grande D, Huo H, Smith RA, Long JA. Community Health Worker Support for Disadvantaged Patients With Multiple Chronic Diseases: A Randomized Clinical Trial. Am J Public Health. 2017;107(10):1660–7. doi:10.2105/AJPH.2017.303985 PubMed PMID: 28817334; PubMed Central PMCID: PMC5607679.

64. Carrasquillo O, Lebron C, Alonzo Y, Li H, Chang A, Kenya S. Effect of a Community Health Worker Intervention Among Latinos With Poorly Controlled Type 2 Diabetes: The Miami Healthy Heart Initiative Randomized Clinical Trial. JAMA Intern Med. 2017 01;177(7):948–54. doi:10.1001/jamainternmed.2017.0926 PubMed PMID: 28459925; PubMed Central PMCID: PMC5818804.

65. Philis-Tsimikas A, Fortmann A, Lleva-Ocana L, Walker C, Gallo LC. Peer-led diabetes education programs in high-risk Mexican Americans improve glycemic control compared with standard approaches: a Project Dulce promotora randomized trial. Diabetes Care. 2011 Sep;34(9):1926–31. doi:10.2337/dc10-2081 PubMed PMID: 21775748; PubMed Central PMCID: PMC3161298.

66. Hughes MM, Yang E, Ramanathan D, Benjamins MR. Community-Based Diabetes Community Health Worker Intervention in an Underserved Chicago Population. J Community Health. 2016 Dec;41(6):1249–56. doi:10.1007/s10900-016-0212-8 PubMed PMID: 27286843.

67. Berkowitz SA, Delahanty LM, Terranova J, Steiner B, Ruazol MP, Singh R, et al. Medically Tailored Meal Delivery for Diabetes Patients with Food Insecurity: a Randomized Cross-over Trial. J Gen Intern Med. 2019 Mar;34(3):396–404. doi:10.1007/s11606-018-4716-z PubMed PMID: 30421335; PubMed Central PMCID: PMC6420590.

68. Seligman HK, Smith M, Rosenmoss S, Marshall MB, Waxman E. Comprehensive Diabetes Self-Management Support From Food Banks: A Randomized Controlled Trial. Am J Public Health. 2018 Jul 19;e1–8. doi:10.2105/AJPH.2018.304528

69. Harnack L, Oakes JM, Elbel B, Beatty T, Rydell S, French S. Effects of Subsidies and Prohibitions on Nutrition in a Food Benefit Program: A Randomized Clinical Trial. JAMA Intern Med. 2016 01;176(11):1610–8. doi:10.1001/jamainternmed.2016.5633 PubMed PMID: 27653735; PubMed Central PMCID: PMC5988257.

70. Olsho LE, Klerman JA, Wilde PE, Bartlett S. Financial incentives increase fruit and vegetable intake among Supplemental Nutrition Assistance Program participants: a randomized controlled trial of the USDA Healthy Incentives Pilot. Am J Clin Nutr. 2016 Aug;104(2):423–35. doi:10.3945/ajcn.115.129320 PubMed PMID: 27334234.

71. HEI Scores for Americans | USDA-FNS [Internet]. [cited 2019 May 31]. Available from: https://www.fns.usda.gov/hei-scores-americans

72. Center for Nutrition Policy and Promotion. Healthy Eating Index (HEI) [Internet]. [cited 2018 Sep 17]. Available from: https://www.cnpp.usda.gov/healthyeatingindex

73. Bryce R, Guajardo C, Ilarraza D, Milgrom N, Pike D, Savoie K, et al. Participation in a farmers' market fruit and vegetable prescription program at a federally qualified health center improves hemoglobin A1C in low income uncontrolled diabetics. Prev Med Rep. 2017 Sep;7:176–9. doi:10.1016/j.pmedr.2017.06.006 PubMed PMID: 28702315; PubMed Central PMCID: PMC5496208.

74. Alley DE, Asomugha CN, Conway PH, Sanghavi DM. Accountable Health Communities--Addressing Social Needs through Medicare and Medicaid. N Engl J Med. 2016 Jan 7;374(1):8–11. doi:10.1056/NEJMp1512532 PubMed PMID: 26731305.

75. Barnidge E, Stenmark S, Seligman H. Clinic-to-Community Models to Address Food Insecurity. JAMA Pediatr. 2017 Jun 1;171(6):507–8. doi:10.1001/jamapediatrics.2017.0067 PubMed PMID: 28384732.

76. Berkowitz SA, Hulberg AC, Standish S, Reznor G, Atlas SJ. Addressing Unmet Basic Resource Needs as Part of Chronic Cardiometabolic Disease Management. JAMA Intern Med. 2017 Feb 1;177(2):244–52. doi:10.1001/jamainternmed.2016.7691 PubMed PMID: 27942709.

77. Centers for Disease Control and Prevention. Measuring Healthy Days [Internet]. CDC, Atlanta, GA; 2000 [cited 2018 Jan 17]. Available from: https://www.cdc.gov/hrqol/pdfs/mhd.pdf

78. Berkowitz SA, Terranova J, Hill C, Ajayi T, Linsky T, Tishler LW, et al. Meal Delivery Programs Reduce The Use Of Costly Health Care In Dually Eligible Medicare And Medicaid Beneficiaries. Health Aff Proj Hope. 2018 Apr;37(4):535–42. doi:10.1377/hlthaff.2017.0999 PubMed PMID: 29608345.

79. Berkowitz SA, Terranova J, Randall L, Cranston K, Waters DB, Hsu J. Association Between Receipt of a Medically Tailored Meal Program and Health Care Use. JAMA Intern Med. 2019 Apr 22. doi:10.1001/jamainternmed.2019.0198 PubMed PMID: 31009050.

80. Gurvey J, Rand K, Daugherty S, Dinger C, Schmeling J, Laverty N. Examining health care costs among MANNA clients and a comparison group. J Prim Care Community Health. 2013 Oct;4(4):311–7. doi:10.1177/2150131913490737 PubMed PMID: 23799677.

81. Palar K, Napoles T, Hufstedler LL, Seligman H, Hecht FM, Madsen K, et al. Comprehensive and Medically Appropriate Food Support Is Associated with Improved HIV and Diabetes Health. J Urban Health Bull N Y Acad Med. 2017 Feb;94(1):87–99. doi:10.1007/s11524-016-0129-7 PubMed PMID: 28097614; PubMed Central PMCID: PMC5359179.

82. Mozaffarian D, Glickman D. Opinion | Our Food Is Killing Too Many of Us. The New York Times [Internet]. 2019 Aug 30 [cited 2019 Sep 10]. Available from: https://www.nytimes.com/2019/08/26/opinion/food-nutrition-health-care.html

83. Medically-Tailored-Meals-Pilot-Program [Internet]. [cited 2019 Sep 11]. Available from: https://www.dhcs.ca.gov/services/ltc/Pages/Medically-Tailored-Meals-Pilot-Program.aspx

84. Verma S. CMS Approves North Carolina's Innovative Medicaid Demonstration To Help Improve Health Outcomes [Internet]. 2018 [cited 2018 Oct 30]. Available from: https://www.healthaffairs.org/do/10.1377/hblog20181024.406020/full/

85. Mass.gov [Internet]. [cited 2019 Sep 11]. MassHealth Demonstration Waiver. Available from: https://www.mass.gov/masshealth-demonstration-waiver

86. United States Department of Agriculture Economic Research Service. Food Security Survey Module [Internet]. [cited 2018 Jan 17]. Available from: https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/survey-tools/

87. Dewitt B, Jalal H, Hanmer J. Computing PROPr Utility Scores for PROMIS® Profile Instruments. Value Health J Int Soc Pharmacoeconomics Outcomes Res. 2020 Mar;23(3):370–8. doi:10.1016/j.jval.2019.09.2752 PubMed PMID: 32197733.

88. Dewitt B, Feeny D, Fischhoff B, Cella D, Hays RD, Hess R, et al. Estimation of a Preference-Based Summary Score for the Patient-Reported Outcomes Measurement Information System: The PROMIS®-Preference (PROPr) Scoring System. Med Decis Mak Int J Soc Med Decis Mak. 2018;38(6):683–98. doi:10.1177/0272989X18776637 PubMed PMID: 29944456; PubMed Central PMCID: PMC6502464.

89. Hanmer J, Cella D, Feeny D, Fischhoff B, Hays RD, Hess R, et al. Selection of key health domains from PROMIS® for a generic preference-based scoring system. Qual Life Res Int J Qual Life Asp

Treat Care Rehabil. 2017;26(12):3377–85. doi:10.1007/s11136-017-1686-2 PubMed PMID: 28825177; PubMed Central PMCID: PMC5682190.

90. Hanmer J, DeWalt DA, Berkowitz SA. Association between Food Insecurity and Health-Related Quality of Life: a Nationally Representative Survey. J Gen Intern Med. 2021 Jan 6. doi:10.1007/s11606-020-06492-9 PubMed PMID: 33409885.

91. Stanulewicz N, Mansell P, Cooke D, Hopkins D, Speight J, Blake H. PAID-11: A brief measure of diabetes distress validated in adults with type 1 diabetes. Diabetes Res Clin Pract. 2019 Mar;149:27–38. doi:10.1016/j.diabres.2019.01.026 PubMed PMID: 30710656.

92. Sarkar U, Karter AJ, Liu JY, Moffet HH, Adler NE, Schillinger D. Hypoglycemia is more common among type 2 diabetes patients with limited health literacy: the Diabetes Study of Northern California (DISTANCE). J Gen Intern Med. 2010 Sep;25(9):962–8. doi:10.1007/s11606-010-1389-7 PubMed PMID: 20480249; PubMed Central PMCID: PMC2917655.

93. Aberegg ES, Collins KK, Hinderliter JM, Smalling SL, Fung TT, Gizlice Z, et al. Validity and Reliability of a Brief Dietary Assessment Questionnaire in a Cardiac Rehabilitation Program. J Cardiopulm Rehabil Prev. 2020 Jul;40(4):280–3. doi:10.1097/HCR.0000000000000505 PubMed PMID: 32604257.

94. Kraschnewski JL, Gold AD, Gizlice Z, Johnston LF, Garcia BA, Samuel-Hodge CD, et al. Development and evaluation of a brief questionnaire to assess dietary fat quality in low-income overweight women in the southern United States. J Nutr Educ Behav. 2013 Aug;45(4):355–61. doi:10.1016/j.jneb.2012.10.008 PubMed PMID: 23340242.

95. PRAPARE. NACHC [Internet]. [cited 2019 Aug 20]. Available from: http://www.nachc.org/research-and-data/prapare/

96. Toobert DJ, Hampson SE, Glasgow RE. The summary of diabetes self-care activities measure: results from 7 studies and a revised scale. Diabetes Care. 2000 Jul 1;23(7):943–50. doi:10.2337/diacare.23.7.943 PubMed PMID: 10895844.

97. Mayberry LS, Gonzalez JS, Wallston KA, Kripalani S, Osborn CY. The ARMS-D out performs the SDSCA, but both are reliable, valid, and predict glycemic control. Diabetes Res Clin Pract. 2013 Nov;102(2):96–104. doi:10.1016/j.diabres.2013.09.010 PubMed PMID: 24209600; PubMed Central PMCID: PMC3915929.

98. Morisky DE, Green LW, Levine DM. Concurrent and predictive validity of a self-reported measure of medication adherence. Med Care. 1986 Jan;24(1):67–74. doi:10.1097/00005650-198601000-00007 PubMed PMID: 3945130.

99. Aikens JE, Piette JD. Longitudinal association between medication adherence and glycaemic control in Type 2 diabetes. Diabet Med J Br Diabet Assoc. 2013 Mar;30(3):338–44. doi:10.1111/dme.12046 PubMed PMID: 23075262; PubMed Central PMCID: PMC3567301.

100. Cohen S, Kamarck T, Mermelstein R. A global measure of perceived stress. J Health Soc Behav. 1983;24(4):385–96. doi:10.2307/2136404

101. Wallston KA, Rothman RL, Cherrington A. Psychometric properties of the Perceived Diabetes Self-Management Scale (PDSMS). J Behav Med. 2007 Oct;30(5):395–401. doi:10.1007/s10865-007-9110-y PubMed PMID: 17522972.

102. Hickey ML, Owen SV, Froman RD. Instrument development: cardiac diet and exercise self-efficacy. Nurs Res. 1992 Dec;41(6):347–51. PubMed PMID: 1437584.

103. Bandura A. Social cognitive theory: an agentic perspective. Annu Rev Psychol. 2001;52:1–26. doi:10.1146/annurev.psych.52.1.1 PubMed PMID: 11148297.

104. Bandura A. Social Foundations of Thought & Action, a Social Cognitive Theory. First Printing edition. Prentice Hall; 1986.

105. Drewnowski A, Eichelsdoerfer P. Can Low-Income Americans Afford a Healthy Diet? Nutr Today. 2010 Nov;44(6):246–9. doi:10.1097/NT.0b013e3181c29f79 PubMed PMID: 20368762; PubMed Central PMCID: PMC2847733.

106. Darmon N, Drewnowski A. Contribution of food prices and diet cost to socioeconomic disparities in diet quality and health: a systematic review and analysis. Nutr Rev. 2015 Oct;73(10):643–60. doi:10.1093/nutrit/nuv027 PubMed PMID: 26307238; PubMed Central PMCID: PMC4586446.

107. Bomberg EM, Neuhaus J, Hake MM, Engelhard EM, Seligman HK. Food Preferences and Coping Strategies among Diabetic and Nondiabetic Households Served by US Food Pantries. J Hunger Environ Nutr. 2019;14(1–2):4–17. doi:10.1080/19320248.2018.1512926 PubMed PMID: 31456865; PubMed Central PMCID: PMC6711179.

108. Leung CW, Epel ES, Willett WC, Rimm EB, Laraia BA. Household Food Insecurity Is Positively Associated with Depression among Low-Income Supplemental Nutrition Assistance Program Participants and Income-Eligible Nonparticipants. J Nutr. 2015 Mar 1;145(3):622–7. doi:10.3945/jn.114.199414

109. Arenas DJ, Thomas A, Wang J, DeLisser HM. A Systematic Review and Meta-analysis of Depression, Anxiety, and Sleep Disorders in US Adults with Food Insecurity. J Gen Intern Med. 2019 Aug 5. doi:10.1007/s11606-019-05202-4 PubMed PMID: 31385212.

110. Kinsey EW, Dupuis R, Oberle M, Cannuscio CC, Hillier A. Chronic disease self-management within the monthly benefit cycle of the Supplemental Nutrition Assistance Program. Public Health Nutr.

2019 Aug;22(12):2248–59. doi:10.1017/S1368980019001071 PubMed PMID: 31104648; PubMed Central PMCID: PMC6641996.

111. Delahanty LM, Chang Y, Levy DE, Porneala B, Dushkin A, Bissett L, et al. Design and participant characteristics of a primary care adaptation of the Look AHEAD Lifestyle Intervention for weight loss in type 2 diabetes: The REAL HEALTH-diabetes study. Contemp Clin Trials. 2018 Aug;71:9–17. doi:10.1016/j.cct.2018.05.018 PubMed PMID: 29803816; PubMed Central PMCID: PMC6067988.

112. Berkowitz S, Wexler DJ, Chang Y, Porneala BC, Delahanty LM. 757-P: Intensive Lifestyle Intervention for Diabetes Patients with Food Insecurity: A Preplanned Subgroup Analysis of the REAL HEALTH Randomized Clinical Trial (RCT). Diabetes. 2019 Jun 1;68(Supplement 1):757-P. doi:10.2337/db19-757-P

113. Delahanty LM, Dalton KM, Porneala B, Chang Y, Goldman VM, Levy D, et al. Improving diabetes outcomes through lifestyle change--A randomized controlled trial. Obes Silver Spring Md. 2015 Sep;23(9):1792–9. doi:10.1002/oby.21172 PubMed PMID: 26260043.

114. Moran A, Thorndike A, Franckle R, Boulos R, Doran H, Fulay A, et al. Financial Incentives Increase Purchases Of Fruit And Vegetables Among Lower-Income Households With Children. Health Aff (Millwood). 2019 Sep 1;38(9):1557–66. doi:10.1377/hlthaff.2018.05420

115. Monsivais P, Aggarwal A, Drewnowski A. Time spent on home food preparation and indicators of healthy eating. Am J Prev Med. 2014 Dec;47(6):796–802. doi:10.1016/j.amepre.2014.07.033 PubMed PMID: 25245799; PubMed Central PMCID: PMC4254327.

116. American Diabetes Association. 5. Lifestyle Management: Standards of Medical Care in Diabetes—2019. Diabetes Care. 2019 Jan 1;42(Supplement 1):S46–60. doi:10.2337/dc19-S005 PubMed PMID: 30559231.

117. Azadbakht L, Fard NRP, Karimi M, Baghaei MH, Surkan PJ, Rahimi M, et al. Effects of the Dietary Approaches to Stop Hypertension (DASH) Eating Plan on Cardiovascular Risks Among Type 2 Diabetic Patients: A randomized crossover clinical trial. Diabetes Care. 2011 Jan 1;34(1):55–7. doi:10.2337/dc10-0676 PubMed PMID: 20843978.

118. Hilliard M, Riekert K, Ockene J, Pbert L, editors. The Handbook of Health Behavior Change, Fifth Edition. 5 edition. New York, NY: Springer Publishing Company; 2018. 574 p.

119. Delahanty LM, Peyrot M, Shrader PJ, Williamson DA, Meigs JB, Nathan DM, et al. Pretreatment, psychological, and behavioral predictors of weight outcomes among lifestyle intervention participants in the Diabetes Prevention Program (DPP). Diabetes Care. 2013 Jan;36(1):34–40. doi:10.2337/dc12-0733 PubMed PMID: 23129133; PubMed Central PMCID: PMC3526204.

120. Kiernan M, Brown SD, Schoffman DE, Lee K, King AC, Taylor CB, et al. Promoting Healthy Weight With "Stability Skills First": A Randomized Trial. J Consult Clin Psychol. 2013 Apr;81(2):336–46. doi:10.1037/a0030544 PubMed PMID: 23106759; PubMed Central PMCID: PMC4428307.

121. Consort - Welcome to the CONSORT Website [Internet]. [cited 2021 Mar 29]. Available from: http://www.consort-statement.org/

122. ICMJE | Home [Internet]. [cited 2021 Mar 29]. Available from: http://www.icmje.org/

123. Wasserstein RL, Lazar NA. The ASA Statement on p-Values: Context, Process, and Purpose. Am Stat. 2016 Apr 2;70(2):129–33. doi:10.1080/00031305.2016.1154108

124. Wasserstein RL, Schirm AL, Lazar NA. Moving to a World Beyond "p < 0.05." Am Stat. 2019 Mar 29;73(sup1):1–19. doi:10.1080/00031305.2019.1583913

125. Amrhein V, Greenland S, McShane B. Scientists rise up against statistical significance. Nature. 2019 Mar;567(7748):7748. doi:10.1038/d41586-019-00857-9

126. It's time to talk about ditching statistical significance. Nature. 2019 Mar 20;567(7748):7748. doi:10.1038/d41586-019-00874-8

127. Inzucchi Silvio E., Kosiborod Mikhail, Fitchett David, Wanner Christoph, Hehnke Uwe, Kaspers Stefan, et al. Improvement in Cardiovascular Outcomes With Empagliflozin Is Independent of Glycemic Control. Circulation. 2018 Oct 23;138(17):1904–7. doi:10.1161/CIRCULATIONAHA.118.035759

128. Davies MJ, D'Alessio DA, Fradkin J, Kernan WN, Mathieu C, Mingrone G, et al. Management of Hyperglycemia in Type 2 Diabetes, 2018. A Consensus Report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). Diabetes Care. 2018 Dec 1;41(12):2669–701. doi:10.2337/dci18-0033 PubMed PMID: 30291106.

129. U.S. Food and Drug AdministrationResearch. Diabetes Mellitus: Developing Drugs and Therapeutic Biologics for Treatment and Prevention [Internet]. 2019 [cited 2019 Aug 7]. Available from: http://www.fda.gov/regulatory-information/search-fda-guidance-documents/diabetes-mellitus-developing-drugs-and-therapeutic-biologics-treatment-and-prevention

130. Tukey JW. Exploratory Data Analysis. 1 edition. Reading, Mass: Pearson; 1977. 712 p.

131. Coffman CJ, Edelman D, Woolson RF. To condition or not condition? Analysing 'change' in longitudinal randomised controlled trials. BMJ Open. 2016 Dec 30;6(12). doi:10.1136/bmjopen-2016-013096 PubMed PMID: 28039292; PubMed Central PMCID: PMC5223669.

132. Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. Biometrics. 2010 Sep;66(3):891–6. doi:10.1111/j.1541-0420.2009.01332.x PubMed PMID: 19764951.

133. Permutt T. Do Covariates Change the Estimand? Stat Biopharm Res. 2020 Jan 2;12(1):45–53. doi:10.1080/19466315.2019.1647874

134. White IR, Horton NJ, Carpenter J, Pocock SJ. Strategy for intention to treat analysis in randomised trials with missing outcome data. BMJ. 2011 Feb 7;342:d40. doi:10.1136/bmj.d40 PubMed PMID: 21300711; PubMed Central PMCID: PMC3230114.

135. Hernán MA, Hernández-Díaz S. Beyond the intention-to-treat in comparative effectiveness research. Clin Trials. 2012 Feb 1;9(1):48–55. doi:10.1177/1740774511420743

136. Hernán MA, Robins JM. Per-Protocol Analyses of Pragmatic Trials. N Engl J Med. 2017 Oct 5;377(14):1391–8. doi:10.1056/NEJMsm1605385 PubMed PMID: 28976864.

137. Little DB, Rubin R. Statistical Analysis with Missing Data. 2nd ed. Wiley-Interscience; 2002.

138. Molenberghs G, Kenward M. Missing Data in Clinical Studies. 1 edition. Chichester: Wiley; 2007. 526 p.

139. Benjamini Y, Hochberg Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. J R Stat Soc Ser B Methodol. 1995;57(1):289–300. Located at: JSTOR

140. Storey JD, Taylor JE, Siegmund D. Strong control, conservative point estimation and simultaneous conservative consistency of false discovery rates: a unified approach. J R Stat Soc Ser B Stat Methodol. 2004 Feb 1;66(1):187–205. doi:10.1111/j.1467-9868.2004.00439.x

141. Valeri L, Vanderweele TJ. Mediation analysis allowing for exposure-mediator interactions and causal interpretation: theoretical assumptions and implementation with SAS and SPSS macros. Psychol Methods. 2013 Jun;18(2):137–50. doi:10.1037/a0031034 PubMed PMID: 23379553; PubMed Central PMCID: PMC3659198.

142. VanderWeele TJ. Explanation in causal inference: developments in mediation and interaction. Int J Epidemiol. 2016 01;45(6):1904–8. doi:10.1093/ije/dyw277 PubMed PMID: 27864406; PubMed Central PMCID: PMC6373498.

143. VanderWeele TJ, Shrier I. Sufficient Cause Representation of the Four-way Decomposition for Mediation and Interaction. Epidemiol Camb Mass. 2016;27(5):e32-33. doi:10.1097/EDE.0000000000000521 PubMed PMID: 27327019; PubMed Central PMCID: PMC4970933.

144. VanderWeele TJ, Tchetgen Tchetgen EJ. Mediation analysis with time varying exposures and mediators. J R Stat Soc Ser B Stat Methodol. 2017 Jun;79(3):917–38. doi:10.1111/rssb.12194 PubMed PMID: 28824285; PubMed Central PMCID: PMC5560424.

145. Bind MA, VanderWeele TJ, Schwartz JD, Coull BA. Quantile causal mediation analysis allowing longitudinal data. Stat Med. 2017 Nov 20;36(26):4182–95. doi:10.1002/sim.7423 PubMed PMID: 28786129; PubMed Central PMCID: PMC5788575.

146. Lin SH, Young J, Logan R, Tchetgen Tchetgen EJ, VanderWeele TJ. Parametric Mediational g-Formula Approach to Mediation Analysis with Time-varying Exposures, Mediators, and Confounders. Epidemiol Camb Mass. 2017;28(2):266–74. doi:10.1097/EDE.0000000000000609 PubMed PMID: 27984420; PubMed Central PMCID: PMC5285457.

147. VanderWeele TJ, Knol MJ. Interpretation of subgroup analyses in randomized trials: heterogeneity versus secondary interventions. Ann Intern Med. 2011 May 17;154(10):680–3. doi:10.7326/0003-4819-154-10-201105170-00008 PubMed PMID: 21576536; PubMed Central PMCID: PMC3825512.

148. Glaser BG, Strauss AL. The discovery of grounded theory: strategies for qualitative research. 1967.

149. Elo S, Kyngäs H. The qualitative content analysis process. J Adv Nurs. 2008 Apr;62(1):107–15. doi:10.1111/j.1365-2648.2007.04569.x PubMed PMID: 18352969.

150. Miles MB, Huberman AM. Qualitative Data Analysis: An Expanded Sourcebook. 2nd edition. Thousand Oaks: SAGE Publications, Inc; 1994. 352 p.

151. Trochim WMK, Donnelly JP. The Research Methods Knowledge Base, 3rd Edition. 3rd edition. Mason, Ohio: Atomic Dog; 2006. 361 p.

152. Creswell J, Plano Clark V. Chapter 3: Choosing a Mixed Methods Design. In: Designing and Conducting Mixed Methods Research. 2nd ed. Sage; 2011. p. 53–106.

153. Creswell J, Plano Clark V. Chapter 7: Analyzing and interpreting data in mixed methods research. In: Designing and Conducting Mixed Methods Research. 2nd ed. Sage; 2011. p. 53–106.